

Session 5.1

Text as Economic Data: From Documents to Counts

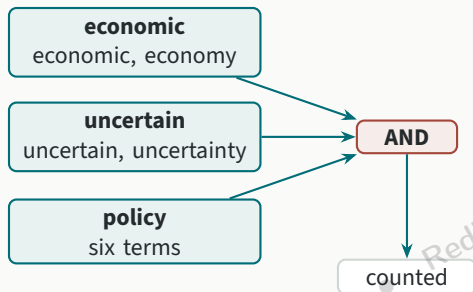
Clean, tokenize, count: how a document becomes a vector a regression can use

Based on Chapter 13 of Miao & Feng, *AI for Economists*



Zhigang Feng

Reading Policy Uncertainty Out of Newspapers

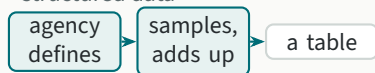


an article counts only if it hits all three groups

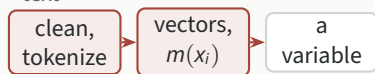
- ▶ Baker, Bloom & Davis (2016) read policy uncertainty from ten US newspapers, month by month since 1985.
- ▶ With c_{pt} hits among N_{pt} articles of paper p in month t : $U_t = \frac{1}{10} \sum_p \frac{c_{pt}/N_{pt}}{\sigma_p}$, and the index is $100 U_t / \bar{U}$.
- ▶ The six policy words were chosen against 12,009 hand-coded articles. The same recipe gives a China index from 1949 (Davis, Liu & Sheng 2019).
- ▶ **No agency publishes this number: it was built from words.**

With Text, the Measurement Step Is Yours

structured data



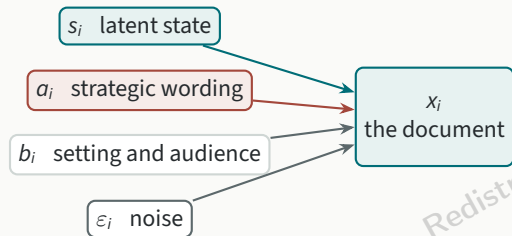
text



the same regression at the end; who did the measuring differs

- ▶ Prices and output arrive measured; FOMC statements, 10-Ks, earnings calls and job ads arrive as language.
- ▶ For a price index the agency defines, samples and adds up. For text nobody has done it.
- ▶ A text variable can be a measure, an outcome or a treatment; this session builds the first.
- ▶ **With text, the researcher does the measuring.**

What a Text Measure Measures



two firms, the same risk s_i : one discloses,
one defects, so x_i differs through a_i

- ▶ The document is not the state you want: $x_i = g(s_i, a_i, b_i) + \varepsilon_i$.
- ▶ The measure $m(x_i)$ enters an ordinary regression: $y_i = \beta m(x_i) + \gamma^\top \mathbf{c}_i + u_i$.
- ▶ If wording a_i and the error u_i share a driver, $\hat{\beta}$ is biased; a bigger corpus does not repair it.
- ▶ **The error moves with the state, so its sign is unknown: ask why $m(x_i)$ is exogenous.**

The Pipeline: Four Steps From a String to a Vector



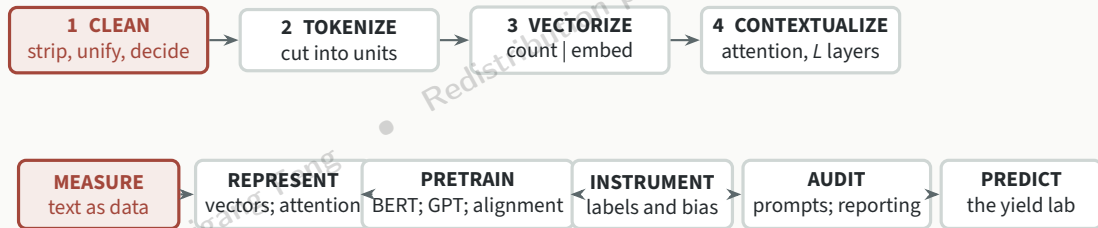
	after step 1: clean	after step 2: tokenize	after step 3: count
EN	The central bank raised interest rates amid rising inflation.	The · central · bank · raised · interest · rates · amid · rising · inflation · .	one row of V counts, mostly 0
ZH	央行上调利率以遏制通胀。	央行 · 上调 · 利率 · 以 · 遏制 · 通胀 · 。	one row of V counts, mostly 0

- ▶ Cleaning and tokenizing fix which strings count as one feature, the coordinates every later model works in. Vectorizing turns tokens into numbers: this deck counts; Deck 5.2 embeds and adds context.
- ▶ **Different choices at any step give different regressors from the same documents.**

Session 5.1

Step 1 • Clean

What to remove, what to unify, and what must stay



Cleaning, Worked on One Sentence

English, scraped from a web page

`<p>The central bank raised ...</p>`

↓ strip mark-up and entities

The central bank raised ...

Chinese, traditional characters

`<p>央行上調利率以遏制通脹。</p>`

↓ 繁 → 简: 調 → 调, 脹 → 胀

央行上调利率以遏制通胀。

C P I 同比上漲 2 . 3 %

↓ 全角 → 半角, 漲 → 涨

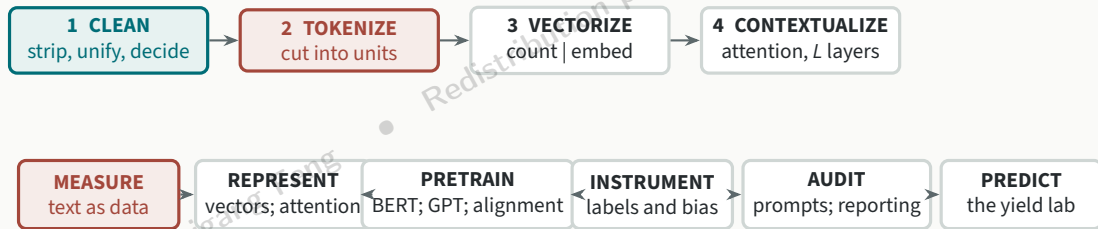
CPI 同比上涨 2.3%

- ▶ Strip what is not text: mark-up, page headers and footers, table debris, broken encodings.
- ▶ Unify what should be one string. Otherwise 通脹 and 通胀 are two words, and C P I and CPI never match.
- ▶ Decide by task: lower-casing merges the Fed with the verb *fed*; in finance %, ? and quotation marks carry information.
- ▶ **Write the rules as a script, and archive it with the data.**

Session 5.1

Step 2 • Tokenize

Cut the string into the units you will count



Tokens and the Vocabulary

English: spaces do most of the cutting

GDP rose by 2 %

Chinese: the boundaries must be inferred

央行 决定 下调 存款准备金率

央行决定下调存款准备金率: the central bank decides to cut the reserve requirement ratio

- ▶ A **corpus** is a collection of texts: a year of FOMC minutes, every annual report in an industry, 一个季度的财经新闻 (a quarter of financial news).
- ▶ A **token** is the smallest unit you count, and **tokenization** cuts a character string into tokens. The **vocabulary** is the set of distinct tokens.
- ▶ **Change the cut and you change every count that follows.**

EN The central bank raised interest rates amid rising inflation .

ZH 央行 上调 利率 以 遏制 通胀 。

the running pair, cut into words

Chinese Segmentation Is a Labelling Problem

研究生生命科学

研究 生命科学

study | life science

研究生 命科学

graduate student | ?

行业政策支持

行业政策 支持

head word: support

行业 政策支持

head word: policy support

one label per character: B begins a word, E ends it

央 行 上 调 利 率

B E B E B E

- ▶ Chinese characters run together, so segmentation is **sequence labelling**: for each character, decide whether a word ends after it.
- ▶ 研究生命科学 reads as *study life science*, or as nonsense if the cut falls after 生.
- ▶ 行业政策支持 splits two sensible ways with different head words: support for an industry policy, or policy support for an industry.
- ▶ **A wrong boundary changes what the sentence is about.**

Domain Compounds Need a User Dictionary

货币政策委员会, the monetary policy committee
general dictionary

货币 政策委员会 head word lost

with a user dictionary

货币政策委员会 kept whole

entries to add first:

社会融资规模 宏观审慎监管框架

- ▶ Segmenters such as Jieba, THULAC and LTP combine a dictionary with a statistical or neural labeller.
- ▶ Domain compounds are missing from general dictionaries, and a wrong cut changes the head word.
- ▶ The fix is a **user dictionary** listing terms such as 社会融资规模 (aggregate financing), so they stay whole.
- ▶ **Add the domain terms before you count, and publish the list.**

Stop Words: Drop the Glue, Keep the Signal

usually dropped

the is of 的 了 是

keep: they carry the signal

not up down more less

因此 therefore 然而 however

- ▶ **Stop words** are frequent function words with little topical content: the, is, of; 的, 了, 是.
- ▶ General lists also delete up, down, more and less, the direction of a macro signal; deleting *not* flips a sentence. 因此 and 然而 carry the logic.
- ▶ Domain matters too: a general sentiment list marks *liability* negative, but in a set of accounts it is a neutral line item (Loughran & McDonald 2011).
- ▶ **A general stop list can delete the signal itself.**

Stems, Lemmas, and a Language Without Inflection

stemming: suffixes cut by rule

investing, invests, invested → invest

studies → studi

lemmatization: a dictionary form

better → good

were → be

Chinese: the verb never changes

已 上调 将 上调

- ▶ **Stemming** chops suffixes by rule, so investing, invests and invested all become *invest*; it also turns studies into *studi*, a non-word.
- ▶ **Lemmatization** returns the dictionary form using the part of speech: better → good, were → be.
- ▶ 上调 (raise) never changes form: tense sits in separate words, 已上调 (has raised) and 将上调 (will raise).
- ▶ **English must choose stem or lemma; Chinese has no need for either.**

n-Grams: One Boundary Can Flip the Sign

scheme A: one token at a time

not sufficient 并未 放缓

“sufficient” and 放缓 (slowing) count on their own, so the sentence’s sign can invert

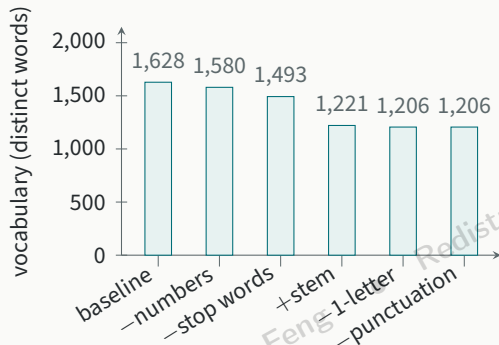
scheme B: adjacent pairs

not_sufficient 并未_放缓

each negation survives as one feature

- ▶ An **n-gram** treats n adjacent tokens as one feature: unigrams are single tokens, bigrams adjacent pairs.
- ▶ Pairs keep compounds and negations whole: central_bank; 货币政策 (monetary policy); 并未_放缓 (has not slowed).
- ▶ Under unigrams a hawkish “not sufficient” can read as reassuring; Hassan et al. (2019) count word pairs to measure political risk.
- ▶ **Pairs keep the relation; the price is a much larger vocabulary.**

Six Pipelines, Five Vocabularies



The last two bars coincide: this tokenizer never emits punctuation, so there are 5 vocabularies, not six.

- ▶ The lab's corpus: 245 FOMC statements, 1994–2026; the 149 from 1994–2014 are used for fitting.
- ▶ Six defensible pipelines on those, each adding one step: 1,628 $\rightarrow$ 1,206 words (–26%).
- ▶ “...to 5-1/4 percent” loses the rate itself; “has moderated *but* remains elevated” loses its turn: infl · moderat · remain · elevat.
- ▶ **Report your pipeline; show the result survives a change** (Denny & Spirling 2018).

Why Subwords

a word split into reusable pieces

macroeconomics → macro economics

not among the lab's 1,628 words, yet
spelled by its byte-pair pieces:

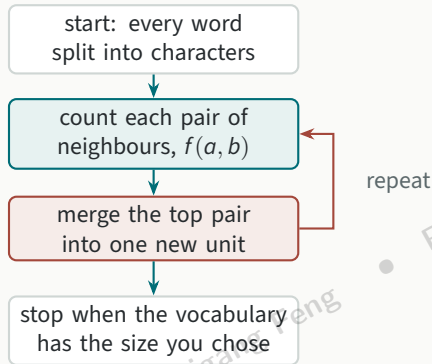
di·s·infl·a·tion·ar·y disinflationary

a policy term coined after the dictionary:

专精特新 → 专 精 特 新

- ▶ Word-level tokenizers fail twice: a word never seen in training gets no representation, and the vocabulary grows with every new form.
- ▶ **Subword tokenization** keeps reusable fragments instead, so a new word is spelled from known pieces.
- ▶ Firm names, tickers and new policy terms such as 专精特新 or 稳健中性 are exactly the words a dictionary misses.
- ▶ **With byte-level subwords, every string has a representation.**

Byte-Pair Encoding: Learn the Pieces From Counts



- ▶ Goal: a vocabulary of a chosen size; frequent strings are one token, rare ones are spelled from pieces (Sennrich, Haddow & Birch 2016).
- ▶ Start from the characters. Count each adjacent pair, $f(a, b) = \sum_w n_w c_{ab}(w)$: word w 's count times the pair's occurrences in w .
- ▶ Merge $(a^*, b^*) = \arg \max f(a, b)$ into a new unit, rewrite every word, count again.
- ▶ **The output is an ordered list of merges. To cut any word, replay them in order.**

BPE Worked: Four Merges on Four Words

a four-word corpus: 通胀 inflation · 通缩 deflation · 通胀率 inflation rate · 利率 interest rate

word	count	start	merge 1	merge 2	merge 3	merge 4
			通+胀, $f = 8$	利+率, $f = 4$	通胀+率, $f = 3$	通+缩, $f = 2$
通胀	5	通 胀	通胀	通胀	通胀	通胀
通缩	2	通 缩	通 缩	通 缩	通 缩	通缩
通胀率	3	通 胀 率	通胀 率	通胀 率	通胀率	通胀率
利率	4	利 率	利 率	利率	利率	利率

- ▶ Merge 1, 通+胀: 5 times in 通胀 plus 3 in 通胀率, $f = 8$. Each column counts again on the rewritten words.
- ▶ A new word reuses the merges: 通缩率 (deflation rate) $\rightarrow$ 通缩 · 率; after two merges 通缩 is still 通 · 缩.
- ▶ On the FOMC statements, 200 merges give un·emplo·y·ment. **Frequency, not meaning, decides the cuts.**

Three Ways to Learn the Pieces

one phrase, “disinflationary pressures eased”, as three tokenizers cut it										pieces
byte-pair (GPT-4’s cl100k)	dis	in	flation	ary	▮	pressures	▮	eased		6
WordPiece (multilingual BERT)	dis	##in	##f	##lation	##ary	pressure	##s	ea	##sed	9
SentencePiece (XLM-R)	_dis	inf	lation	ary	_pressure	s	_ea	sed		8

scheme	how a piece enters the vocabulary	word boundary	used by
byte-pair (Sennrich et al. 2016)	merge the most frequent pair, on bytes	space inside the piece	GPT-2 to 4, LLaMA 3
WordPiece (Schuster & Nakajima 2012)	merge the pair that most raises likelihood	## on later pieces	BERT (Devlin et al. 2019)
SentencePiece unigram (Kudo & Richardson 2018)	prune a large candidate set by likelihood	_ on a first piece	T5, XLM-R

► Each learned its pieces on its own corpus: one phrase, three cuts.

One Sentence, Five Tokenizers

the Chinese sentence, final 。 included, as each tokenizer cuts it

tokens: ZH / EN

cl100k (GPT-4)	E5 A4 AE 行 上 调 利 率 以 E9 81 8F 制 通 E8 83 80 。	15 / 10
o200k (GPT-4o)	央 行 上 调 利 率 以 E9 81 8F 制 通 E8 83 80 。	14 / 10
BERT, English	[UNK] 行 上 [UNK] [UNK] [UNK] [UNK] [UNK] [UNK] [UNK] [UNK] [UNK] 。	12 / 10
BERT, Chinese	央 行 上 调 利 率 以 遏 制 通 胀 。	12 / 20
BERT, multilingual	央 行 上 调 利 率 以 遏 制 通 胀 。	12 / 11

- ▶ Byte-level BPE splits a character it saw rarely into raw UTF-8 bytes (red, in hex): 央 is three bytes, cut in two by cl100k. The newer o200k keeps 央 whole but still splits 遏 and 胀.
- ▶ English BERT knows few Chinese characters: 9 of its 12 tokens are [UNK]. Chinese BERT cuts every character, and cuts the English sentence into 20 pieces.
- ▶ **The same sentence is 10 tokens to one model and 20 to another.**

The Tokenizer Belongs to the Model

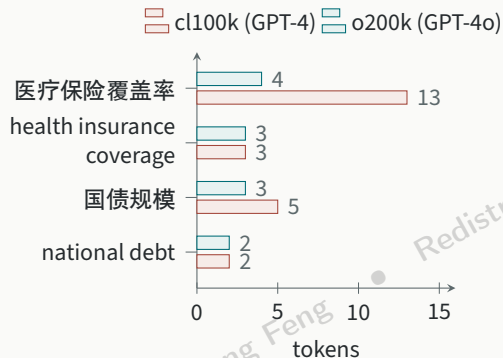
社会融资规模, cut four ways

o200k	社会	融资	规模					
cl100k	社	会	E8	9E	8D	资	规	模
BERT, Chinese	社	会	融	资	规	模		
BERT, English	社	会	[UNK]	[UNK]	[UNK]	[UNK]		

aggregate financing to the real economy; in
cl100k, 融 is three raw bytes

- ▶ One term, four cuts: 3 pieces, 8, 6 characters, or [UNK]s. “GDP rose by 2%”, 5 tokens cut by words, is 6 or 7 GPT tokens.
- ▶ A model’s first layer is an **embedding table**: one learned vector per token ID. Feed it another tokenizer’s IDs and every row is the wrong vector.
- ▶ Mixed Chinese–English notes and filings that mix 简 and 繁 call for byte-level BPE.
- ▶ **Report the tokenizer, and its version, as you report the sample period.**

Translate, or Read in the Original?

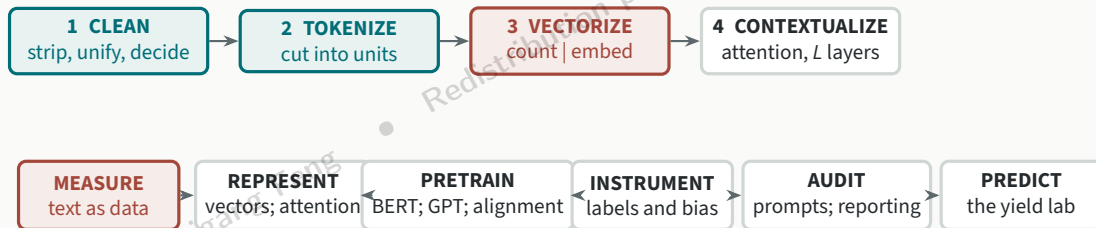


- ▶ One idea need not map one to one: 稳健 means both stable and sound, while “prudent” keeps only caution; the colour of 适度宽松 is hard to render.
- ▶ Translating drifts the colour; a multilingual model keeps the wording, with quality that varies by language.
- ▶ Chinese costs more tokens (left), so it fills a model’s window sooner.
- ▶ **If a conclusion rests on a hard-to-translate phrase, report both routes.**

Session 5.1

Step 3 • Count

Turn tokens into vectors: one axis per word



From Tokens to Numbers: IDs and One-Hot Vectors

vocabulary, numbered

	fed	cut	hike	support	curb	growth	inflation	respond
ID	1	2	3	4	5	6	7	8
$\mathbf{o}_{\text{fed}}$	1	0	0	0	0	0	0	0
$\mathbf{o}_{\text{hike}}$	0	0	1	0	0	0	0	0

$\mathbf{o}_{\text{fed}}^T \mathbf{o}_{\text{hike}} = 0$: no two different words share a coordinate

- ▶ Number the vocabulary: each distinct token gets an ID $v = 1, \dots, V$.
- ▶ Token v becomes the **one-hot vector** $\mathbf{o}_v \in \{0, 1\}^V$: a 1 in place v , 0 everywhere else.
- ▶ For two different words, $\mathbf{o}_v^T \mathbf{o}_{v'} = 0$: every pair of words is **orthogonal** by construction, whatever the words mean. *hike* and *raise* are as far apart as *hike* and *cut*.
- ▶ **A document is the sum of its one-hot vectors: a row of counts.**

Bag of Words and the Document-Term Matrix

	fed	cut	hike	support	curb	growth	inflation	respond
$\mathbf{z}_1$	1	1	0	1	0	1	0	0
$\mathbf{z}_2$	1	0	1	0	1	0	1	0
$\mathbf{z}_3$	1	0	1	0	0	0	2	1

three documents, $V = 8$: $\mathbf{Z} \in \mathbb{R}^{3 \times 8}$,
frequency counts

- ▶ Three short documents, already cleaned and tokenized: $d_1 = \text{fed cut support growth}$; $d_2 = \text{fed hike curb inflation}$; $d_3 = \text{fed hike respond inflation inflation}$.
- ▶ Count each word and ignore order: the **document-term matrix** $\mathbf{Z} \in \mathbb{R}^{D \times V}$, one row per document.
- ▶ Binary counts record presence (inflation twice in d_3 gives 1); frequency counts give 2.
- ▶ **Each row is its document's vector $\mathbf{z}_i$.**

TF-IDF, Worked Through

term	df_v	$idf_v = \log(D/df_v)$	$tf_{v,3}$	TF-IDF in document 3
fed	3	$\log(3/3) = 0$	1	0
hike	2	$\log(3/2) = 0.405$	1	0.405
inflation	2	$\log(3/2) = 0.405$	2	0.811
respond	1	$\log(3/1) = 1.099$	1	1.099

- ▶ $TF\text{-}IDF_{v,i} = tf_{v,i} \times \log(D/df_v)$, with $tf_{v,i}$ the count of word v in document i and df_v the number of documents containing it: frequent here, rare across the corpus. d_3 becomes $(0, 0, 0.405, 0, 0, 0, 0.811, 1.099)$.
- ▶ A graded stop list: boilerplate such as “financial statements” has $df \approx D$ and a weight near 0, with no list written in advance.
- ▶ **The trap: if every statement contains “not”, its weight goes to zero too.**

The Bag Forgets Word Order

GDP growth exceeded inflation

inflation exceeded GDP growth

both become the same count vector:

GDP	growth	exceeded	inflation
1	1	1	1

the rate cut supports growth

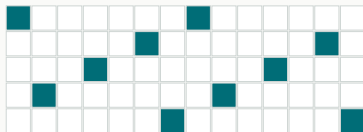
growth supports the rate cut

the same bag

- ▶ “GDP growth exceeded inflation” and “inflation exceeded GDP growth” use the same words, so they get the same vector.
- ▶ “The rate cut supports growth” and “growth supports the rate cut” state opposite causal directions, in identical bags.
- ▶ Harmless when you count topic words, as the uncertainty index does; fatal when direction, cause or negation is the object.
- ▶ **Word order is gone. Deck 5.2 puts it back with position vectors.**

Sparse: Most of the Matrix Is Zero

a document-term matrix, schematic



filled cells are non-zero; in the lab's 245
× 1,163 TF-IDF matrix, 85.5% of the
cells are 0

- ▶ In the three-document example, each row has 4 non-zero entries out of 8. With tens of thousands of words, often fewer than one cell in a hundred is non-zero.
- ▶ With fewer documents than words ($D \ll V$), OLS on all V columns is not identified: $\mathbf{Z}^\top \mathbf{Z}$ cannot be inverted.
- ▶ Reduce the dimension, with topic models, or penalise, with the Lasso.
- ▶ **That is why text regressions borrow the tools of machine learning.**

Comparing Documents: Cosine on Counts

	<i>fed</i>	<i>cut</i>	<i>hike</i>	<i>support</i>	<i>curb</i>	<i>growth</i>	<i>inflation</i>	<i>respond</i>	<i>raise</i>
d_1	1	1	0	1	0	1	0	0	0
d_2	1	0	1	0	1	0	1	0	0
d_3	1	0	1	0	0	0	2	1	0
d'_2	1	0	0	0	1	0	1	0	1

$$\cos(d_1, d_2) = 1 / (2 \cdot 2) = 0.25$$

$$\cos(d_2, d_3) = (1 + 1 + 2) / (2\sqrt{7}) = 0.756$$

$$\cos(d'_2, d_3) = (1 + 2) / (2\sqrt{7}) = 0.567, d'_2 \text{ is } d_2 \text{ with } \textit{raise} \text{ for } \textit{hike}$$

- ▶ Words are orthogonal; documents are not. A row is a sum of one-hots, so in the **cosine similarity** $\mathbf{z}_i^\top \mathbf{z}_j / (\|\mathbf{z}_i\| \|\mathbf{z}_j\|)$ every word two documents share adds to the numerator.
- ▶ d_1, d_2 share only *fed*: 0.25 on counts; 0 under TF-IDF, where *fed* weighs 0.
- ▶ Swap *hike* for *raise* in d_2 : same meaning, but the cosine with d_3 falls from 0.756 to 0.567.
- ▶ **Counts see shared strings, not shared meaning; Deck 5.2 repairs this.**

Topic Models: Each Document Is a Mixture of Topics



d_3 with $\theta_3 = (0.1, 0.9)$, one draw:

hike inflation respond inflation fed
 $z = 2$ $z = 2$ $z = 2$ $z = 2$ $z = 1$

- **Latent Dirichlet allocation** (Blei, Ng & Jordan 2003) models how the counts were written. Topic k is a distribution over words, $\phi_k \sim \text{Dir}_V(b)$.
- Document i has **topic shares** $\theta_i \sim \text{Dir}_K(a)$, summing to 1; a small a means few topics per document.
- Each token: a topic $z \sim \text{Cat}(\theta_i)$, then a word $w \sim \text{Cat}(\phi_z)$. Only the words are seen.
- **A document mixes topics; word order plays no part.**

Fitting LDA: Collapsed Gibbs Sampling, Step by Step

toy state. $n_k^{(v)}$: tokens of word v in topic k

	fed	cut	hike	support	curb	growth	inflation	respond	
$k = 1$	1	1	0	1	0	1	0	0	$n_1^{(\cdot)} = 4$
$k = 2$	2	0	2	0	1	0	3	1	$n_2^{(\cdot)} = 9$

$n_i^{(k)}$: tokens of document i in topic k

d_1	4	0	d_2	0	4	d_3	0	5
-------	---	---	-------	---	---	-------	---	---

Why it works: draws of each topic given all the others form a Markov chain that settles on the posterior. “Collapsed”: ϕ, θ are integrated out; only the z ’s are drawn.

1. Give every token a random topic z ; fill the tables.
2. Take one token (word v , document i) out of the counts. For each k :

$$p_k \propto \frac{n_k^{(v)} + b}{n_k^{(\cdot)} + Vb} \times (n_i^{(k)} + a)$$

($a, b > 0$ keep every p_k positive).

3. Draw its new topic from p ; put it back.
4. Every token once is a **sweep**; run hundreds. Read $\hat{\phi}$ and $\hat{\theta}$ off the final counts.

Fitting LDA, Worked: One Token, Then Three Seeds

Step 2 on *fed* in d_3 ($a = 0.1, b = 0.01, V = 8$).
 Taking it out: $n_2^{(\text{fed})} 2 \rightarrow 1, n_2^{(\cdot)} 9 \rightarrow 8, n_3^{(2)} 5 \rightarrow 4$.

	likes <i>fed</i>	used by d_3	p_k
topic 1	$1.01/4.08 = 0.248$	$0 + 0.1$	0.05
topic 2	$1.01/8.08 = 0.125$	$4 + 0.1$	0.95

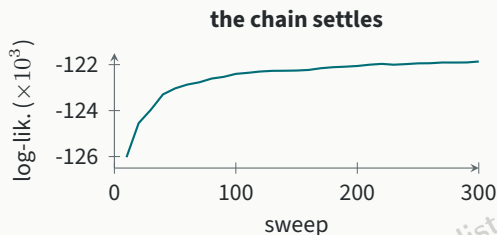
Topic 1 likes *fed* more, but d_3 uses only topic 2, so *fed* almost surely goes back there.

after 200 sweeps, three seeds; $\hat{\theta}_{i1}$ is document i 's share of topic 1

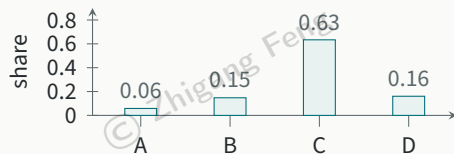
seed	topic 1	topic 2	$\hat{\theta}_{11}$	$\hat{\theta}_{21}$	$\hat{\theta}_{31}$
42	fed, inflation, hike	cut, growth, curb	0.50	0.98	0.98
2	cut, growth, support	fed, inflation, hike	0.74	0.02	0.02
3	inflation, fed, hike	cut, fed, growth	0.02	0.98	0.98

- ▶ Every seed puts d_2 and d_3 , the hikes, in one topic and the cut words in the other: unprompted, “tightening” and “easing”. d_1 's mix changes with the seed, and topic numbers swap (seed 3).
- ▶ **Three documents are too few for a stable fit; the FOMC run has 21,673 tokens.**

Lab A5: Topics in the FOMC Statements, the Setup

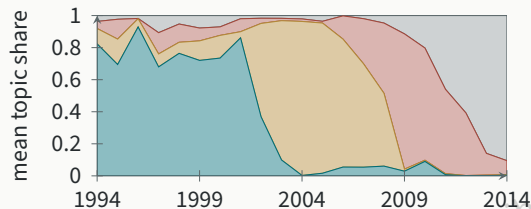


one row of $\hat{\theta}$: the statement of 2008-12-16



- ▶ The question: what did the 1994–2014 statements talk about? Can 4 shares replace 719 counts?
- ▶ The data: 149 statements, lower-cased; no punctuation, numbers, stop words or words under 3 letters; words in ≥ 5 statements and $\leq 80\%$: 719 words, 21,673 tokens.
- ▶ The fit: the sampler above, $K = 4$, $a = 0.1$, $b = 0.01$, 300 sweeps, about a minute.
- ▶ **The output: $\hat{\phi}$, four word lists, and $\hat{\theta}$, a 149×4 table of shares.**

LDA on the FOMC Statements: What It Found

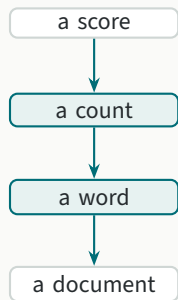


top words, and the name we gave

- reserve, discount, board, available
A: discount-rate announcements
- chairman, risks, decided, voting
B: the vote and the balance of risks
- financial, markets, levels, housing
C: crisis, markets and housing
- longer, term, consistent, pace
D: longer-run goals and purchases

- ▶ Drafting eras as much as policy themes: A before 2002, when statements reported discount-rate approvals; B from the vote roster of March 2002; C 2008–10; D 2011–14.
- ▶ In A5: name the topics before seeing the plot, compare with a neighbour, refit with seed 7 or $K = 6$. $\hat{\theta}_i$ is estimated, so its error enters any regression.
- ▶ **The model finds words that occur together; we chose K and the names.**

When Counting Still Wins



every coordinate is a word, so a score can be traced back to its source

- ▶ When the question is how often defined words occur, transparency is the value: the uncertainty index, finance sentiment lists, political risk.
- ▶ An audit traces a score back to its words and documents; a dense vector cannot.
- ▶ Cosines of binary word vectors of 10-K product descriptions define product-market rivals (Hoberg & Phillips 2016); the same can be tried on the business sections of other annual reports.
- ▶ **Under about a thousand documents, TF-IDF is the sound baseline.**

No Peeking: Build the Weights From Past Statements Only

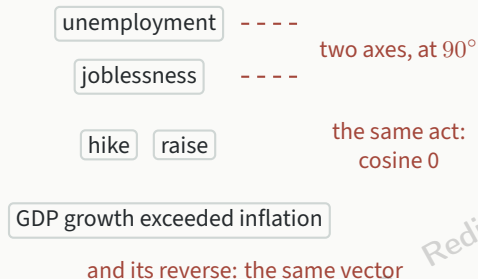
1994–2014: 149 statements 2015–2026

a 2010 statement 2020

	fit on 1994–2014		fit on all 245	
word	df	IDF	df	IDF
balance	33	1.51	89	1.01
securities	49	1.11	120	0.71
pandemic	0	no column	19	2.56

- ▶ IDF counts documents, $\log(D/df_v)$, and which documents you count is a choice.
- ▶ Count all 245: *balance* is in 56 of the 96 later statements, so it weighs 1.01, not 1.51, in a 2010 row. That row now depends on 2020: **look-ahead**.
- ▶ The rule: fit the vocabulary and IDF on 1994–2014 and apply them unchanged. *pandemic* gets no column; Deck 5.6 measures the cost.
- ▶ **A weight fitted on the future cannot be reproduced in real time.**

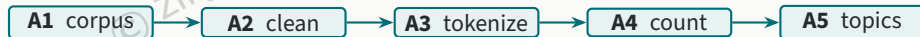
What Counts Cannot See



- ▶ Cleaning, tokenizing and counting turned each document into a row of counts: transparent, auditable, and blind to meaning.
- ▶ Synonyms sit on perpendicular axes, word order is gone, and every word is its own axis.
- ▶ Deck 5.2 gives up one axis per word: words become points, and nearness means similarity.
- ▶ **Counts tell you which words were used, not what was meant.**

Lab, Part A: Clean, Tokenize and Count the FOMC Statements

cell	what you run	the question you answer
A1	load 245 statements; set the 1994–2014 window	How many statements may a fitted object see?
A2	five cleaning switches, one at a time	Which switch moves the vocabulary most, and what does it delete?
A3	byte-pair encoding, 200 merges	How is a word the window never saw, such as <i>reflationary</i> , cut?
A4	document–term matrix and TF-IDF, fitted on the window	How much held-back text falls outside the vocabulary? What would a full-sample IDF change?
A5	LDA with $K = 4$, collapsed Gibbs	Your names for the topics, a neighbour's, and what changes with seed 7 or $K = 6$?



► Every number on the counting slides is one you can reproduce.

Session 5.2

Meaning in Vectors: Embeddings, Attention, the Transformer

Embed, then contextualize: from one vector per word to one vector per word in its sentence

Based on Chapter 13 of Miao & Feng, *AI for Economists*

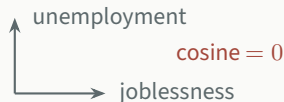


Zhigang Feng

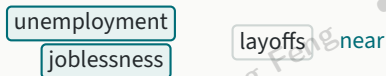
Where the Pipeline Stands



counting: one axis per word



embedding: one point per word

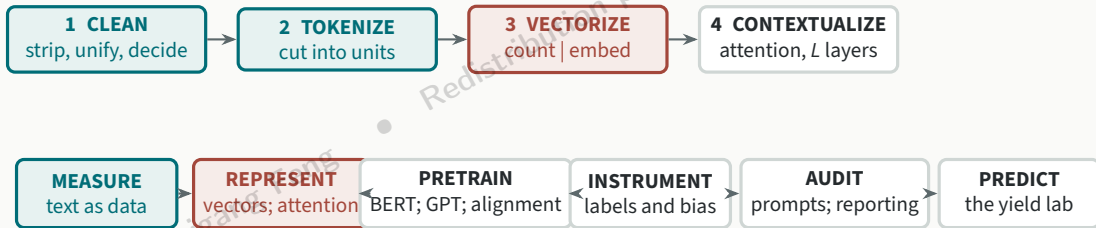


- ▶ Deck 5.1 cleaned the text, cut it into **tokens** (the units you count) and counted them: each document became a **vector**, one entry per vocabulary word.
- ▶ Every word is its own axis, so two different words have inner product 0 and **cosine** (the cosine of their angle) 0: *unemployment* and *joblessness*, *hike* and *raise*.
- ▶ **The repair: make words points, where nearness means similar meaning.**

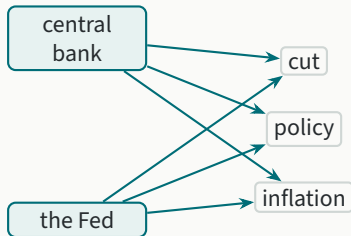
Session 5.2

Step 3 • Embed

Words become points, and closeness becomes meaning



Know a Word by the Company It Keeps



two words, one set of neighbours:
their vectors land close

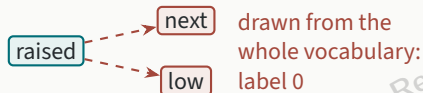
- ▶ Firth (1957): “You shall know a word by the company it keeps.” This is the **distributional hypothesis**.
- ▶ *Central bank* and *Fed* both appear near *cut*, *policy* and *inflation*. Nobody declared them related; their contexts did.
- ▶ A **distributed representation** gives each word w a dense vector $\mathbf{u}_w \in \mathbb{R}^d$ with d far below the vocabulary size V , learned from co-occurrence. Its axes carry no names.
- ▶ **Similar context profiles, nearby vectors: the orthogonality of counting is gone.**

Word2Vec, Step 1: A Window Turns Text Into Pairs

$\Delta = 2$ around the centre word



four real pairs: label 1

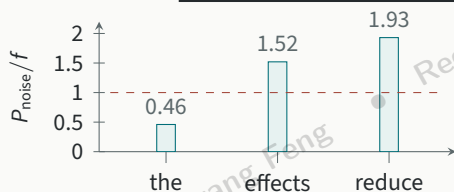


$K = 2$ noise words for the pair (raised, central)

- ▶ Each token in turn is the **centre word**; tokens within Δ positions are its **context words**: each pair is real, label 1.
- ▶ Per real pair, K **noise words** are drawn from the whole vocabulary and paired with the same centre: label 0.
- ▶ $P_{\text{noise}}(w) = f(w)^{3/4} / \sum_v f(v)^{3/4}$, $f(w)$ = w 's share of the tokens. By f , the commonest words dominate; uniform, rare ones do; $3/4$ worked best (Mikolov et al. 2013).
- ▶ **The corpus supplies the data: real pairs against random ones.**

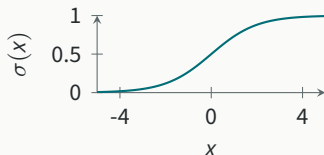
Example: The Training Rows for One Sentence

real pair	label	its $K = 2$ noise pairs, drawn at random	label
(raised, central)	1	(raised, next), (raised, low)	0
(raised, bank)	1	(raised, lending), (raised, 3)	0
(raised, interest)	1	(raised, in), (raised, less)	0
(raised, rates)	1	(raised, susan), (raised, earlier)	0



- ▶ Real draws (seed 42) from the 624 words seen ≥ 10 times in the 1994–2014 statements.
- ▶ *the* is 7.34% of their tokens but 3.35% of draws; *reduce*, seen 10 times, is drawn 1.93 times its share.
- ▶ **12 labelled rows from one centre word, every label free.**

Word2Vec, Step 2: The Objective

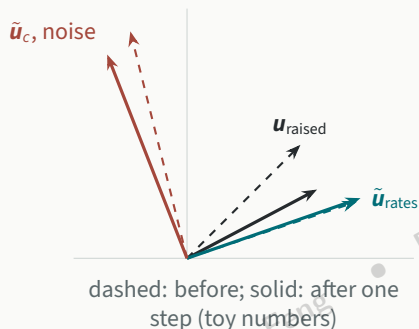


- ▶ $\sigma(x) = 1/(1 + e^{-x})$: $\sigma(\tilde{\mathbf{u}}_c^\top \mathbf{u}_w)$ is the model's probability that (w, c) is a real pair.
- ▶ Each word has a centre vector $\mathbf{u}_w$ and a context vector $\tilde{\mathbf{u}}_w$.

$$\max_{\{\mathbf{u}, \tilde{\mathbf{u}}\}} \sum_{t=1}^T \sum_{\substack{-\Delta \leq j \leq \Delta \\ j \neq 0}} \left[\log \sigma(\tilde{\mathbf{u}}_{w_{t+j}}^\top \mathbf{u}_{w_t}) + \sum_{k=1}^K \log \sigma(-\tilde{\mathbf{u}}_{c_k}^\top \mathbf{u}_{w_t}) \right], \quad c_k \sim P_{\text{noise}}$$

- ▶ Outer sums: every position t of the corpus and every offset j in the window, so every real pair once. Inner sum: that pair's K noise words.
- ▶ A **binary logit** on the last slide's rows: $\log \sigma(x)$ scores a 1, $\log \sigma(-x) = \log(1 - \sigma(x))$ a 0; the vectors are the parameters.
- ▶ **Skip-gram with negative sampling (SGNS): real pairs up, random pairs down.**

Word2Vec, Worked: One Gradient Step Pulls and Pushes



- ▶ A real pair (raised, rates) and a noise pair (raised, c), in 2-D: $x^+ = \tilde{\mathbf{u}}_{\text{rates}}^\top \mathbf{u}_{\text{raised}} = 2.0$, $\sigma(x^+) = 0.88$; $x^- = 1.5$, $\sigma(-x^-) = 0.18$.
- ▶ Gradient for the centre:
 $(1 - \sigma(x^+)) \tilde{\mathbf{u}}_{\text{rates}} - \sigma(x^-) \tilde{\mathbf{u}}_c$, a pull of 0.12, a push of 0.82. The context vectors move too.
- ▶ One step, $\eta = 0.25$: $x^+ 2.0 \rightarrow 2.08$, $x^- 1.5 \rightarrow 0.28$; objective $-1.83 \rightarrow -0.96$.
- ▶ **Words with the same contexts get the same pulls, so they end up pointing the same way.**

Window, Dimension, Domain

small window: syntax

quantitative → easing

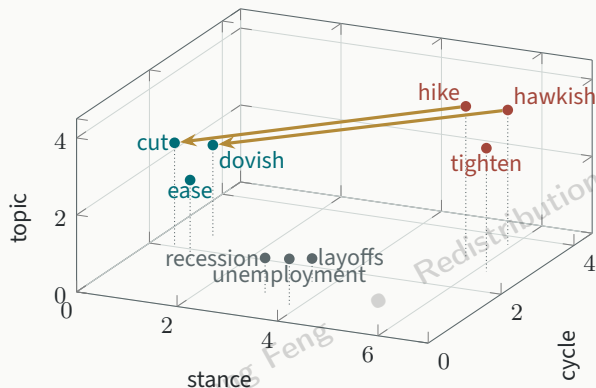
large window: topic

quantitative easing → asset purchases
balance sheet

related by subject, not by grammar

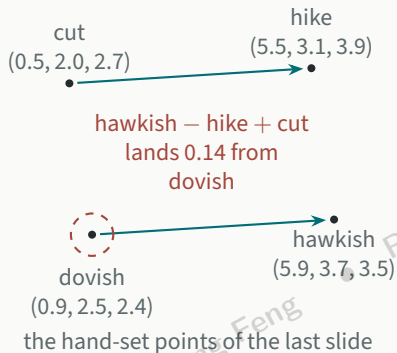
- ▶ A small window ($\Delta \leq 3$) finds syntax; a large one ($\Delta \geq 10$) finds topic, usually what a measure wants. d is typically 100–300.
- ▶ Domain matters more. Trained on general text, *discount* sits near shop markdowns, not the discount rate; *yield* near harvests, not bond yields.
- ▶ CBOW predicts the centre from the averaged context, diluting rare words; GloVe fits $\mathbf{u}_v^\top \tilde{\mathbf{u}}_{v'} + b_v + \tilde{b}_{v'} \approx \log N_{vv'}$ (Pennington et al. 2014).
- ▶ **Train on the corpus you measure.**

The Geometry: Clusters, Parallel Moves, Coupled Axes



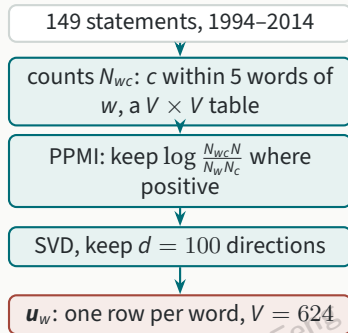
- Nine points, hand-set on named axes; real vectors have 100–300 unnamed ones.
- **Clusters:** tightening (red), easing (teal), the cycle (grey).
- **Parallel moves:** hike→cut and hawkish→dovish point the same way (gold).
- **Coupling:** *hike* is high on stance *and* topic.
- **A direction can mean something; an axis rarely does.**

Analogies, Worked, and Why the Metric Is Cosine



- ▶ **Parallelogram rule:** $\mathbf{u}_A - \mathbf{u}_B \approx \mathbf{u}_C - \mathbf{u}_D$.
Here hike - cut = (5.0, 1.1, 1.2) and hawkish - dovish = (5.0, 1.2, 1.1): cosine 0.9996.
- ▶ hawkish - hike + cut = (0.9, 2.6, 2.3), 0.14 from dovish. The famous case: king - man + woman $\approx$ queen.
- ▶ $\|\mathbf{u} - \mathbf{u}'\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{u}'\|^2 - 2\|\mathbf{u}\|\|\mathbf{u}'\|\cos\theta$: distance also reads the lengths, and length tracks frequency.
- ▶ **Compare directions with cosine, not lengths with distance.**

Lab B1: Word Vectors From the FOMC Statements, the Setup



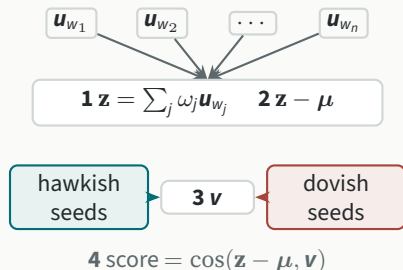
- ▶ The question: what do the Committee's own words mean, judged by their company in its statements? Vectors trained on news would answer for news.
- ▶ 149 statements are too few for skip-gram, so lab cell B1 counts, on words seen ≥ 10 times.
- ▶ PMI: how much more often w and c meet than chance (N : all pairs). Then $\text{PPMI} \approx \mathbf{U}_d \Sigma_d \mathbf{V}_d^\top$; $\mathbf{u}_w = \text{row } w \text{ of } \mathbf{U}_d \Sigma_d^{1/2}$, length 1.
- ▶ **A close cousin of SGNS, whose optimum has $\tilde{\mathbf{u}}_c^\top \mathbf{u}_w = \text{PMI}(w, c) - \log K$ (Levy & Goldberg 2014).**

What the FOMC Vectors Learned

word	its five nearest neighbours by cosine				
inflation	expectations	longer	pressures	term	stable
employment	maximum	objectives	seeks	foster	toward
committee	today	decided	open	raise	particular
accommodation	remove	can	removed	begin	take

- ▶ Read the neighbours as phrases: *inflation* meets *expectations*, *longer* and *pressures* in “longer-term inflation expectations” and “inflation pressures”; *maximum employment* is the mandate’s. A 5-word window learns set phrases.
- ▶ “raise : raised :: lower : ?” has no answer: *raised* occurs fewer than 10 times and gets no vector — the method, not the geometry.
- ▶ In B1: print other words’ neighbours; change the window or minimum count; see what survives.
- ▶ **Read the neighbours before trusting a direction.**

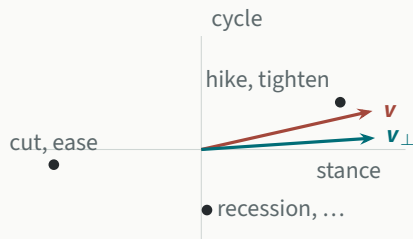
From Word Vectors to a Document Score



document, hand-set points	uncentred	centred
hike, tighten	+0.94	+0.97
cut, ease	+0.54	-0.96

- ▶ Goal: one number per document on a concept (how hawkish?), counting synonyms no list names. ω_j : uniform or TF-IDF, fixed before reading.
- ▶ Centre first: all vectors share a common part, so uncentred even “cut, ease” scores +0.54.
- ▶ Used for emotion vs reason in Congress (Gennaro & Ash 2022), evidence vs intuition (Aroyehun et al. 2025); see Ash & Hansen (2023).
- ▶ **The score is a regressor: validate it like one.**

What a Direction Bundles In



centred documents and directions,
seen on the stance–cycle plane

- ▶ “Recession, unemployment, layoffs” has no policy word, yet scores -0.25 : dovish.
- ▶ Why: hikes are written about in booms, cuts in recessions, so $\mathbf{v} = (5.03, 1.13, 1.13)$ points up the cycle axis too. King and queen differ in one dimension; hike and cut in two.
- ▶ Repair: remove a cycle direction $\mathbf{c}$ (policy minus cycle words), $\mathbf{v}_{\perp} = \mathbf{v} - (\mathbf{v}^{\top} \hat{\mathbf{c}}) \hat{\mathbf{c}}$: scores $+0.88, -1.00, 0.00$.
- ▶ **Remove what you do not mean. The lab’s FOMC hawk–dove score (B2–B3) tracks the policy path at only $+0.07$; centred, $+0.05$.**

Four Limits of One Vector per Word

Polysemy. *rate*: the policy rate, an exchange rate, the inflation rate, a growth rate; *discount*: the discount rate or a price cut. One vector averages every sense.

Corpus dependence. *Mask* sat near face and costume before 2020, near mandate and protection after. A regime shift in language breaks a time series.

Negation. Averaging keeps “growth is slowing” next to “growth has not slowed” — the turn forward guidance is about.

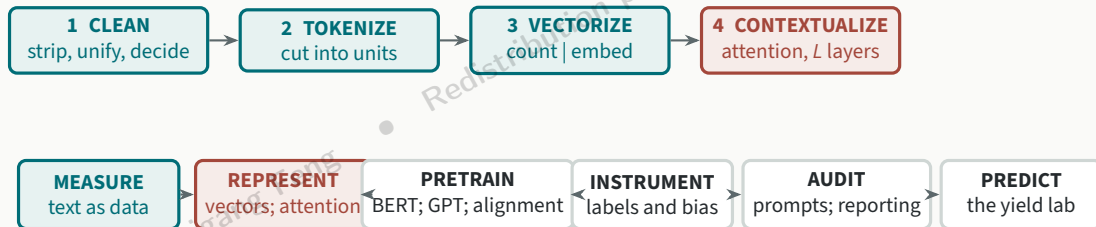
Inherited bias. man : computer programmer :: woman : homemaker (Bolukbasi et al. 2016); association effects at human size (Caliskan et al. 2017). Noise in a feature, a finding as an object.

- All four come from one vector per word, whatever the sentence around it.

Session 5.2

Step 4 • Contextualize

A weight computed from the sentence, instead of fixed before it is read



One Word, Two Sentences

(A) Fed raised policy **rate** inflation

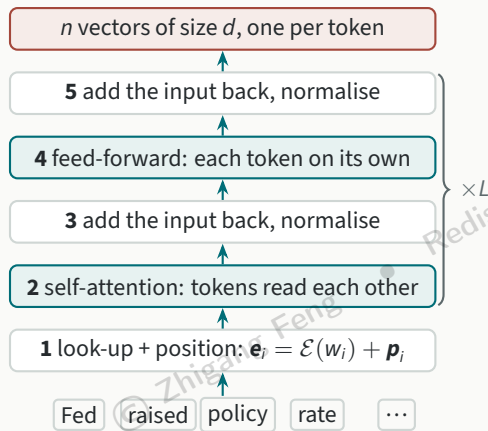
the policy instrument

(B) unemployment **rate** fell

a labour-market statistic

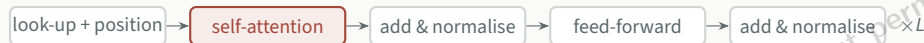
- ▶ (A) The Fed raised its policy *rate* in response to persistent inflation. (B) The unemployment *rate* fell to its lowest level in decades.
- ▶ The same token, *rate*, names two economic objects: an instrument the Fed sets, and a labour-market statistic.
- ▶ A static vector gives both the same $\mathbf{u}_{\text{rate}}$, and a document vector with fixed ω cannot separate them.
- ▶ **The weights must be computed from the sentence itself.**

The Transformer Layer at a Glance



- In: n tokens. Out: n vectors of size d , each now carrying its sentence.
- Only step 2 moves information between positions; the others work on one token's vector.
- Steps 3 and 5 keep each vector and rescale it; step 4 transforms it with one small shared network.
- **Steps 2–5 repeat L times (12 in multilingual BERT). The next frames open each box.**

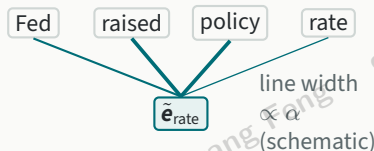
The Idea: Re-Weight the Sentence for Each Word



embedding table $\mathcal{E}$: one row per token ID

... rate

... raised



- ▶ The **embedding table** $\mathcal{E}$ holds one learned vector per token ID; add a position vector: $\mathbf{e}_i = \mathcal{E}(w_i) + \mathbf{p}_i$.
- ▶ Rebuild token i from the sentence: $\tilde{\mathbf{e}}_i = \sum_j \alpha_{ij} \mathcal{E}(w_j)$, $\alpha_{ij} \geq 0$, $\sum_j \alpha_{ij} = 1$. The next frames refine this.
- ▶ Unlike ω , the α_{ij} come from *this* sentence: in (A) *rate* leans on raised and policy; in (B) on unemployment and fell.
- ▶ **The same token, different neighbours read, a different vector: self-attention.**

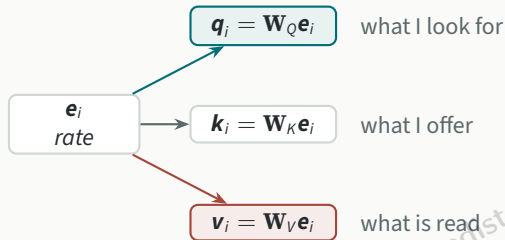
Attention Began as Alignment in Translation

	央行	上调	利率
The	0.70	0.15	0.15
central bank	0.85	0.10	0.05
raised	0.10	0.80	0.10
rates	0.05	0.15	0.80

rows: the English word being written; columns: the Chinese source (central bank, raised, interest rate). Illustrative weights

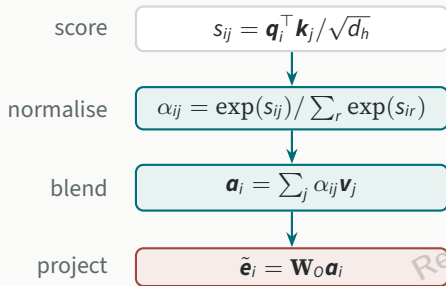
- ▶ A recurrent network reads one token at a time into one state; translating, it squeezed the whole source into it.
- ▶ Attention let each word being written look back at every source word (Bahdanau et al. 2015): *raised* reads 上调, *rates* reads 利率. This is **cross-attention**.
- ▶ The Transformer drops the recurrence (Vaswani et al. 2017): it trains in parallel, and any two positions are one step apart.
- ▶ **Self-attention is this alignment, turned on one sentence.**

Query, Key, Value: Three Roles for One Token



- ▶ Three learned matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_h \times d}$ give each token three roles — request, index label, content: a soft database look-up.
- ▶ In (A), the query of *rate* meets the keys of Fed, raised, policy and inflation.
- ▶ In “The discount rates applied to pension funds were revised”, *rates* meets discount and pension instead.
- ▶ **Three separate matrices, so attention is asymmetric: i may read j heavily while j barely reads i .**

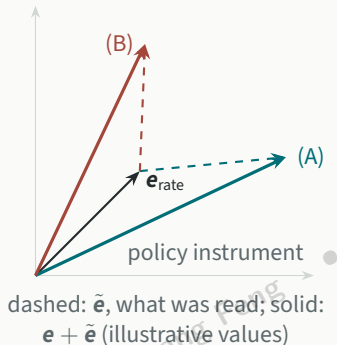
Score, Normalise, Blend, Project



- Sizes: $\mathbf{q}_i, \mathbf{k}_j, \mathbf{v}_j \in \mathbb{R}^{d_h}$; s_{ij} is one number; $\mathbf{a}_i \in \mathbb{R}^{d_h}$; $\mathbf{W}_O \in \mathbb{R}^{d \times d_h}$, so $\tilde{\mathbf{e}}_i \in \mathbb{R}^d$, the size of $\mathbf{e}_i$.
- The softmax makes each row a distribution: position i spends one unit of attention, on the value vectors.
- Why $\sqrt{d_h}$: the variance of $\mathbf{q}^\top \mathbf{k}$ grows with the head width d_h . Unscaled, the softmax piles onto one position and the gradients elsewhere vanish.
- $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V, \mathbf{W}_O$ and $\mathcal{E}$ are learned; α_{ij} is computed for every input: a share, never a coefficient.

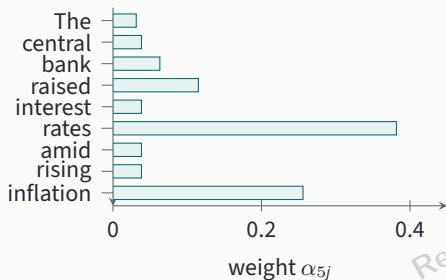
From $\mathbf{e}_i$ to $\tilde{\mathbf{e}}_i$: Same Size, New Content

labour statistic



- ▶ $\mathbf{e}_i$: the token alone, the same in every sentence (for *rate*, between senses).
- ▶ $\tilde{\mathbf{e}}_i$: what it read, a weighted mix of other tokens' values, also of size d , so it adds: $\mathbf{e}_i + \tilde{\mathbf{e}}_i$.
- ▶ One piece of information it carries: *which kind of rate*. (A) reads Fed, raised, policy, inflation: $\tilde{\mathbf{e}} = (0.84, 0.08)$; (B) unemployment, fell, lowest, decades: $(0.03, 0.73)$.
- ▶ **Same size, new content: the two *rate* vectors, identical before, now have cosine 0.77.**

Example 1: Scores to Weights for the Query *interest*



token j	s_{5j}	$e^{s_{5j}}$	α_{5j}
The	-0.2	0.82	0.03
central	0	1.00	0.04
bank	0.5	1.65	0.06
raised	1.1	3.00	0.11
interest	0	1.00	0.04
rates	2.3	9.97	0.38
amid	0	1.00	0.04
rising	0	1.00	0.04
inflation	1.9	6.69	0.26

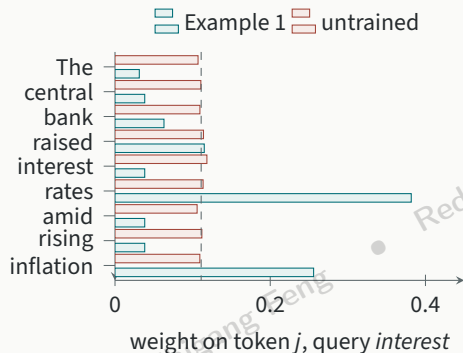
- ▶ Query *interest*. Trained, $s_{5j} = \mathbf{q}_5^\top \mathbf{k}_j / \sqrt{d_h}$; here set for illustration. $\alpha_{5j} = e^{s_{5j}} / \sum_r e^{s_{5r}}$, the sum 26.13.
- ▶ **64% of the attention goes to *rates* and *inflation*: this *interest* is an interest rate, not a stake.**

Example 2: Blend, Project, Add

		level	up	prices
rates	$0.38 \times$	1	0.3	0
inflation	$0.26 \times$	0	0.2	1
raised	$0.11 \times$	0	1	0
rising	$0.04 \times$	0	0.5	0.2
bank	$0.06 \times$	0.5	0	0
$+ \dots = \mathbf{a}_5$		0.46	0.30	0.26
$\mathbf{e}_{\text{interest}}$		1	0	0
$\mathbf{e} + \tilde{\mathbf{e}}$		1.46	0.30	0.26

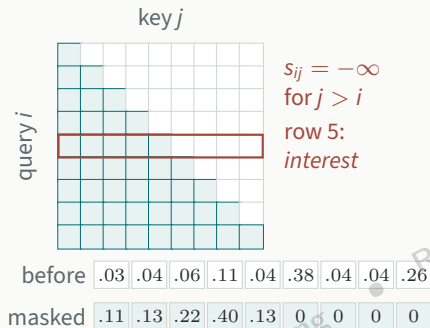
- ▶ Each token's value vector gets three named coordinates: rate level, direction up, price pressure (illustrative; real ones are unnamed).
- ▶ Blend with Example 1's weights:
 $\mathbf{a}_5 = \sum_j \alpha_{5j} \mathbf{v}_j = (0.46, 0.30, 0.26)$;
 with $\mathbf{W}_O = \mathbf{I}$ here, $\tilde{\mathbf{e}}_5 = \mathbf{a}_5$.
- ▶ Add the token's own vector:
 $(1.0, 0.0, 0.0) + (0.46, 0.30, 0.26) = (1.46, 0.30, 0.26)$.
- ▶ **Same size, new content: *interest* now says “going up” and “price pressure”.**

Example 3: The Same Arithmetic, Untrained



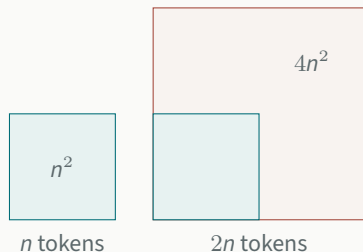
- ▶ The same tokens through an untrained head: random embeddings ($d = 16$) and random $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ ($d_h = 8$).
- ▶ Its row for *interest*: 0.106–0.118, against $1/9 = 0.111$. Nearly flat.
- ▶ Training (Deck 5.3) fits $\mathbf{W}_Q, \mathbf{W}_K$ so that the keys that matter score high. Lab C1 reproduces Example 1; C1b, this head.
- ▶ **The arithmetic is fixed; the pattern is learned.**

The Causal Mask: A Writer Must Not Read Ahead



- ▶ Goal: a generator writes token $i + 1$ from tokens $1, \dots, i$ (Deck 5.3). Trained on whole sentences at once, unmasked, position i could read the answer, token $i + 1$, and copy it.
- ▶ Set $s_{ij} = -\infty$ for $j > i$: then $\alpha_{ij} = 0$ and each row renormalises over $j \leq i$. This is the **causal mask**.
- ▶ Example 1 masked: the sum falls to 7.47; *raised* takes 0.40; *rates* and *inflation* get 0. Lab C1b masks the untrained head.
- ▶ **An encoder keeps the square; a decoder uses the triangle.**

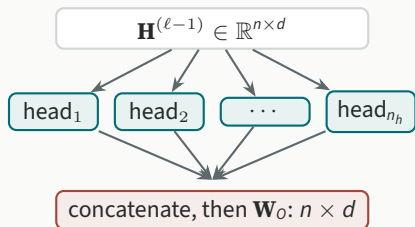
What the Square Costs



text	tokens n	scores n^2
the nine-token sentence	9	81
median FOMC statement	410	168,100
longest statement	1,053	1,108,809
100 statements	41,000	1,681,000,000

- ▶ Every query scores every key: n tokens give n^2 scores in each head of each layer, stored as an $n \times n$ table. (Counts: the lab's 245 statements, GPT-4's tokenizer.)
- ▶ Doubling n quadruples work and memory; 100 times longer costs 10,000 times as much.
- ▶ So a long filing is cut into blocks that cannot see each other, severing a cross-reference from Item 7 back to Item 1A: Deck 5.3's window limit.
- ▶ **The square is the price of letting every token read every other.**

Many Heads, and the Whole Sentence at Once



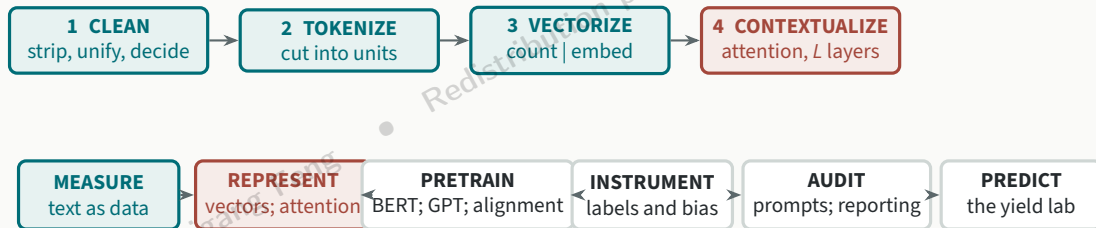
$4n_h d_h d = 4d^2$ parameters, whatever n_h is

- ▶ Stack the tokens as rows, $\mathbf{H}^{(\ell-1)} \in \mathbb{R}^{n \times d}$. Each head computes $\text{softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d_h}) \mathbf{V}$ at width $d_h = d/n_h$; the output is again $n \times d$.
- ▶ Heads are like readers of one set of minutes: one follows inflation language, one the labour market, one references back such as “since the Committee’s last meeting”.
- ▶ More heads redistribute capacity; they add no parameters. Nothing assigns their jobs.
- ▶ **A head weights positions; it does not split a document into topics.**

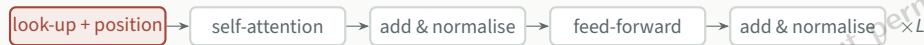
Session 5.2

One Transformer Layer

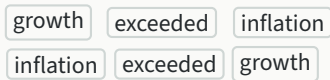
Position, residual connections, normalisation, and the feed-forward block



Attention Cannot See Word Order



the same tokens, opposite claims

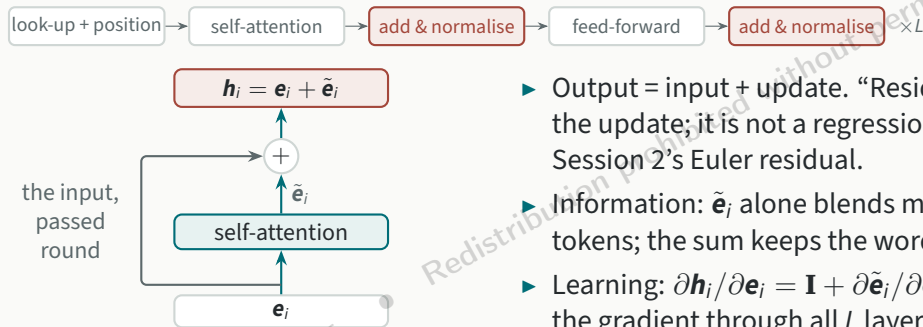


queries and keys see tokens, not places

$$\mathbf{e}_i = \mathcal{E}(w_i) + \mathbf{p}_i$$

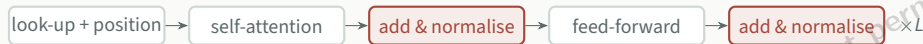
- ▶ Without position vectors, attention is permutation-invariant: shuffle the tokens and no score changes (a causal mask leaks some order; Haviv et al. 2022).
- ▶ **Sinusoidal:** $p_{i,2l} = \sin(i/10^{8l/d})$, $p_{i,2l+1} = \cos(i/10^{8l/d})$.
- ▶ Learned positions stop at the training length; RoPE (Su et al. 2024) with scaling and ALiBi (Press et al. 2022) read past it.
- ▶ **Order is not in the mechanism; it is added to the input.**

Residual Connection: Keep the Token, Add What It Read



- ▶ Output = input + update. “Residual” names the update; it is not a regression residual or Session 2’s Euler residual.
- ▶ Information: $\tilde{e}_i$ alone blends mostly other tokens; the sum keeps the word too.
- ▶ Learning: $\partial h_i / \partial e_i = \mathbf{I} + \partial \tilde{e}_i / \partial e_i$. The $\mathbf{I}$ carries the gradient through all L layers; without it, one Jacobian per layer shrinks it: the **vanishing gradient**.
- ▶ A layer with nothing to add learns $\tilde{e}_i \approx 0$ and passes its input on.

Layer Normalisation: Standardise Each Token's Vector



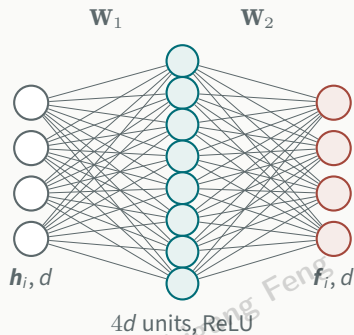
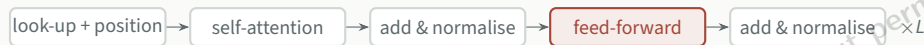
one token's vector, $d = 4$

2	4	6	8
$\downarrow (\mathbf{x} - \mu) / \sigma$ $\mu = 5, \sigma = 2.24$			
-1.34	-0.45	0.45	1.34

then $\gamma \odot (\cdot) + \beta$, learned per coordinate

- ▶ $\text{LN}(\mathbf{x}) = \gamma \odot (\mathbf{x} - \mu) / \sqrt{\sigma^2 + \epsilon} + \beta$: μ, σ = mean and s.d. of the d entries of *one* token's vector; $\gamma, \beta \in \mathbb{R}^d$ learned; ϵ tiny.
- ▶ Why: each residual adds to the vector, so its scale drifts from layer to layer; LN puts each layer's input on one scale, like standardised regressors.
- ▶ Originally applied after the addition; Machine B and most current models apply it before each block.
- ▶ **Neither the residual nor LN moves information between tokens. Only attention does.**

The Feed-Forward Block: Transform Each Token on Its Own



- ▶ Attention only averages values, a linear mix. Combining features — *rate, raised, inflation* into “tightening” — needs a nonlinear step.
- ▶ $f_i = W_2 \text{ReLU}(W_1 h_i + b_1) + b_2$, $W_1 \in \mathbb{R}^{4d \times d}$, $W_2 \in \mathbb{R}^{d \times 4d}$, $\text{ReLU}(x) = \max(0, x)$: Session 2’s one-hidden-layer network.
- ▶ Same weights at every position; output size d , added back. Machine B uses GELU, a smooth ReLU.
- ▶ **Attention moves information between tokens; this block transforms each one.**

Example: A Hidden Unit as a Key-Value Memory

	level	up	prices	tight	
h	1.46	0.30	0.26	0.00	
key 1	1	1	1	0	→ 0.52
value 1	0	0	0	1	
key 2	1	-1	0	0	→ 0.00
$h + f$	1.46	0.30	0.26	0.52	●

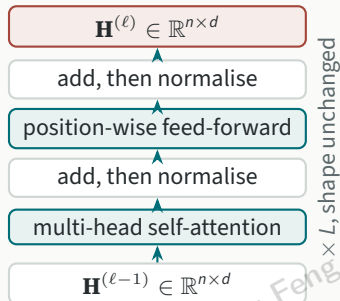
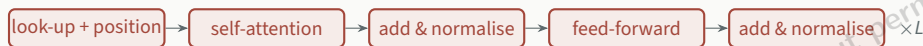
- ▶ Illustrative: coordinates rate level, up, price pressure, tightening; h from Example 2; bias -1.5 . Unit 1's key (a row of W_1): $2.02 + (-1.5) = 0.52$. It fires on “a rate going up with price pressure” and adds $0.52 \times$ its value (a column of W_2).
- ▶ Unit 2 looks for “a rate going down”: -0.34 , ReLU 0, silent. The vector now says “tightening”, which no token said.
- ▶ **Rows of W_1 detect patterns; columns of W_2 write back what goes with them. Trained keys respond to recognisable text patterns (Geva et al. 2021).**

Where the Parameters Are: Machine B, Part by Part

part of Machine B	count	share
feed-forward $\mathbf{W}_1$, 3 layers	198,144	31.4%
feed-forward $\mathbf{W}_2$, 3 layers	196,992	31.2%
attention $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$, 3 layers	148,608	23.5%
attention $\mathbf{W}_O$, 3 layers	49,536	7.8%
position table, 128 positions	16,384	2.6%
character table, 78 characters	9,984	1.6%
output layer, 78 characters	9,984	1.6%
layer norms before attention	768	0.1%
layer norms before feed-forward	768	0.1%
final layer norm	256	0.0%
whole model	631,424	100%

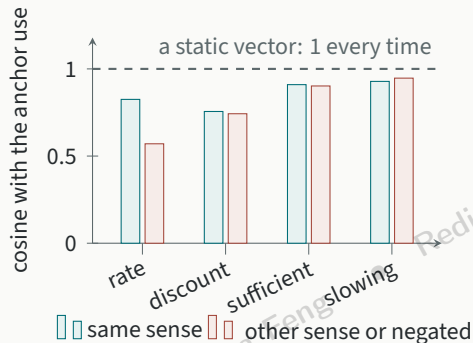
- **Machine B:** Session 1's character-level text generator, trained on the 245 FOMC statements less every 20th; 3 layers, 4 heads, $d = 128$.
- Per layer, feed-forward $8d^2$ against attention's $4d^2$: 62.6% of the model. Lab C2 prints this table.
- **Most parameters sit in the feed-forward blocks, where associations learned from text are kept.**

One Layer, Stacked L Times



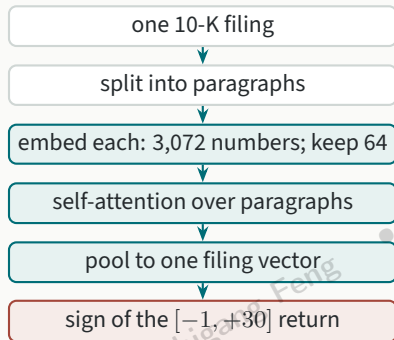
- ▶ The layer's output has the input's shape, $n \times d$, so layers stack with no change of interface.
- ▶ Shallow layers learn to local syntax (*quantitative easing*); deep ones link paragraphs, a “risk factor” early in an annual report with a “regulatory change” later.
- ▶ Machine B stacks 3 layers, $d = 128$; multilingual BERT 12, $d = 768$.
- ▶ **$\mathbf{H}^{(L)}$ is where Deck 5.3 starts.**

What Context Buys, Measured



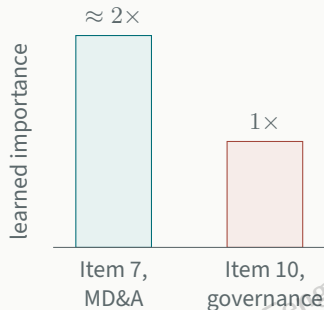
- ▶ Multilingual BERT (12 layers, $d = 768$), last layer: one word in three sentences, cosine with its anchor use.
- ▶ *rate* separates its senses, 0.825 vs 0.570 (unemployment rate); *discount* barely, 0.756 vs 0.743 (a price discount).
- ▶ Negation barely moves it: *sufficient* 0.910 vs 0.902; for *slowing* the negated use is the closer, 0.947 vs 0.928.
- ▶ **Context can fix a sense; whether it does, and whether it reads “not”, must be measured.**

Reading 10-Ks by Paragraph Attention: The Design



- ▶ Attention is not tied to word tokens: Kim, Muhn, Nikolaev & Zhang (2024) embed each paragraph as one vector, so the paragraph is the “token”.
- ▶ 76,929 filings from 2004 to 2024, split into 20,712,462 paragraphs.
- ▶ **The target is a direction: is the cumulative return from one day before the filing to thirty days after positive?**

What the Learned Weights Say, and What They Cannot



- ▶ Attention beats both attention-free baselines (Kim et al. 2024).
- ▶ MD&A is attended most, almost twice as much as governance.
- ▶ Learned to predict returns: “important” means useful, not causal. In a sentence, too, α_{ij} is an intermediate quantity, and boilerplate repeated in every filing may spread it thin.
- ▶ **Attention routes information inside the text it is given. It does not retrieve truth.**

The Whole Pipeline

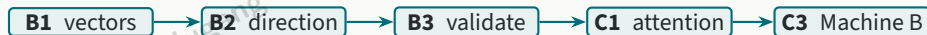


rung	“rates” becomes	size	similar words	senses	you can read
counts, TF-IDF	one column of the matrix	$V \sim 10^4\text{--}10^5$, mostly zeros	orthogonal	one per word	each axis is a word
word vector	one point $\mathbf{u} \in \mathbb{R}^d$	$d = 100\text{--}300$, dense	close	one per word	no axis is a word
contextual vector	a point per sentence	● $d = 768$ in multilingual BERT	close	one per use	the weights, not the meaning

- ▶ Report every step: the cleaning script; the tokenizer and its version; the vector method, corpus and period; the model and its version.
- ▶ **Each rung buys meaning and sells the audit chain: pick the rung the design needs.**

Lab, Parts B and C: Vectors and Attention on the FOMC Statements

cell	what you run	the question you answer
B1	word vectors by PPMI and SVD	Which neighbours are phrases? Which survive another window?
B2	the hawk–dove direction	Which seeds survive, and what does the score bundle in?
B3	validation on the policy path	Where does the embedding score fail?
C1, C1b	Example 1; the untrained head, masked	What does the mask set to 0? How flat is the row?
C2–C3	Machine B's parameters and maps	Where are the parameters? Is the upper triangle 0?



- **Each cell reproduces the slide numbers it names; C4–C5 belong to Deck 5.3.**

Session 5.3

Pretrained Language Models: BERT, GPT, Alignment, and Five Limits

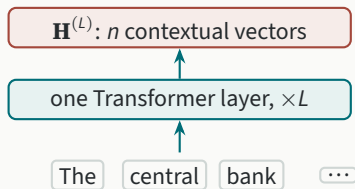
What pretraining buys, what alignment changes, and the five limits that follow from the architecture

Based on Chapter 13 of Miao & Feng, *AI for Economists*



Zhigang Feng

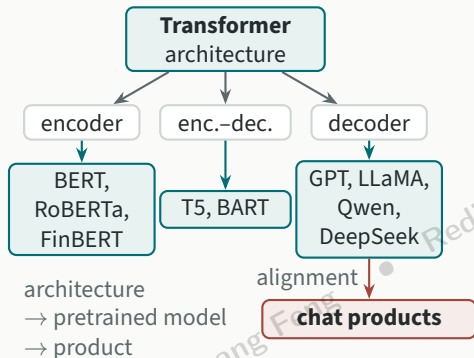
Where the Pipeline Stands



untrained, the attention row for *interest* runs 0.106–0.118: flat

- ▶ Deck 5.2 built one Transformer layer and stacked it L times: n tokens in, n contextual vectors $\mathbf{H}^{(L)}$ out.
- ▶ With random weights it reads nothing: the architecture holds no language yet.
- ▶ This deck: how the weights are fit on raw text; how the two model families measure; how a text generator becomes an assistant; which limits follow.
- ▶ **Pretraining puts language into the weights; how you use them decides the measure.**

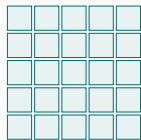
The Family Tree: Architecture, Pretrained Model, Product



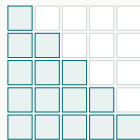
- **Architecture**: the wiring of Deck 5.2's layers, with random weights.
- **Pretrained model**: the same wiring with weights fit on a corpus. Encoders read; decoders write.
- **Product**: a pretrained decoder taught to follow instructions (**alignment**) and served with default settings.
- **Name the layer you mean: the product is not the architecture.**

One Block, Three Wirings

encoder: all
read all



decoder: i reads
 $1..i$



encoder-decoder: an encoder
reads the input; a decoder
writes the output, reading the
encoder by cross-attention

- ▶ On “The central bank raised interest rates amid rising inflation”:
- ▶ **Encoder**: nine tokens in, nine contextual vectors out, each built from both sides.
- ▶ **Decoder**: “The central bank ... amid rising” in; a probability for every vocabulary token as the next one out.
- ▶ **Encoder-decoder**: the sentence in, another sequence out: a translation, or a record {action: raise, reason: inflation}.
- ▶ **The wiring fixes what the model can be asked for.**

Session 5.3

The Encoder Route

BERT: fill in blanks, pool, add a head, adapt to a domain



Pretraining an Encoder: Fill in the Blanks



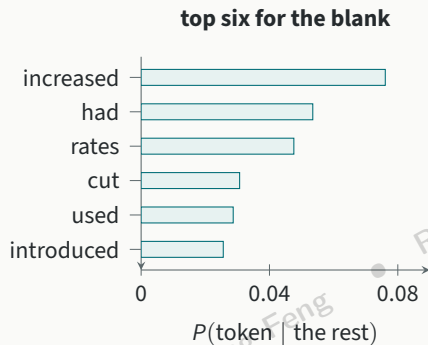
both sides needed: *raised? cut?*

choose 15% of positions; of those
80% $\rightarrow$ [MASK], 10% $\rightarrow$ a random
token, 10% unchanged

BERT also predicted whether sentence B follows A; RoBERTa found that dropping it “matches or slightly improves” results (Liu et al. 2019).

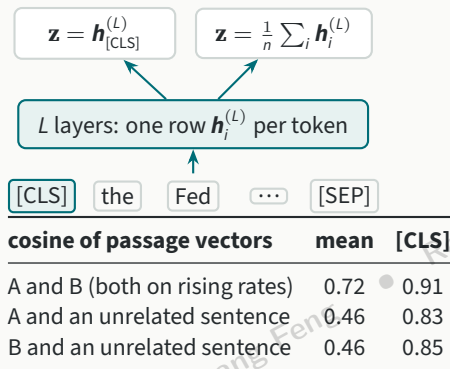
- ▶ Goal: fit the weights on raw text, no labels: hide words and let the text supply the answers.
- ▶ $\mathcal{M}$: the chosen 15% of positions; $\tilde{w}_{1:n}$: the corrupted input. [MASK] never appears in use, hence the 10% kept and 10% swapped (Devlin et al. 2019).
- ▶ Maximise $\sum_{i \in \mathcal{M}} \log P_{\theta}(w_i \mid \tilde{w}_{1:n})$, where $P_{\theta}(\cdot) = \text{softmax}(\mathbf{W}_{\text{out}} \mathbf{h}_i^{(L)})$ over the V tokens and θ is every weight.
- ▶ **To fill a blank the model must read both sides, so $\mathbf{H}^{(L)}$ encodes context.**

Example: A Pretrained Encoder Fills the Blank



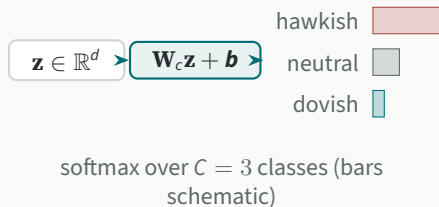
- ▶ Multilingual BERT's masked-word head, recorded offline, on “The central bank [MASK] interest rates amid rising inflation.”: *increased* 0.076, *cut* 0.031, the true *raised* 0.019, so this blank's loss is $-\log 0.019 = 3.97$.
- ▶ “Inflation (unemployment) is rising, so the Committee decided to [MASK] the target range”: $P(\text{raise})$ 0.0014 vs 0.0029, the wrong way round.
- ▶ **Pretraining delivers the kind of word, not the economics: hence domain text and labels.**

From Token Vectors to One Vector: [CLS] and Mean Pooling



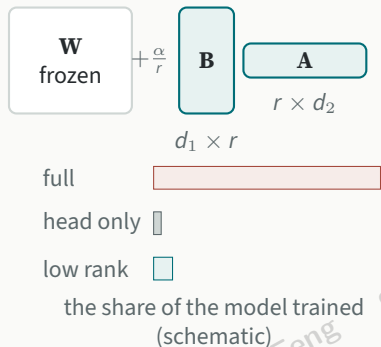
- ▶ Input: [CLS] $w_1 \dots w_n$ [SEP]. [CLS] is a slot with no word whose last-layer row stands for the text; [SEP] marks the end.
- ▶ Two ways to one vector, **pooling**: the [CLS] row, or the mean of all rows.
- ▶ Recorded on multilingual BERT: a policy statement (A) and an annual-report passage (B) score 0.72 by the mean, against 0.46 with unrelated text. Raw [CLS] rows are all alike.
- ▶ **Untuned, pool by the mean; the [CLS] row becomes useful once a task trains it.**

Fine-Tuning: A Classification Head on $\mathbf{z}$



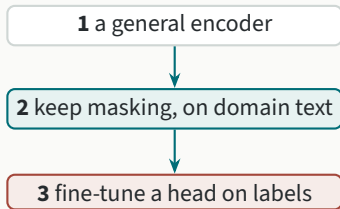
- ▶ Goal: a stance for each sentence, learned from N hand-labelled ones.
- ▶ Head: $\mathbf{p} = \text{softmax}(\mathbf{W}_c \mathbf{z} + \mathbf{b})$, $\mathbf{W}_c \in \mathbb{R}^{C \times d}$; loss $-\frac{1}{N} \sum_i \log p_{i, y_i}$, y_i sentence i 's label.
- ▶ A multinomial logit on the encoder's features; full fine-tuning also moves the encoder.
- ▶ Counts: the head has $C(d + 1) = 2,307$ numbers ($d = 768$); the encoder 177,853,440.
- ▶ **A few hundred labels can pin down the head, not the encoder.**

Three Ways to Fine-Tune



- **Full**: update all 177,853,440 weights. Closest fit; with few labels it overfits and loses the general representation.
- **Frozen encoder**: train the head alone. Cheap; the representation cannot adapt.
- **Low-rank adaptation** (LoRA; Hu et al. 2022): $\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r} \mathbf{B} \mathbf{A}$, $\mathbf{A}$ random, $\mathbf{B} = \mathbf{0}$ at the start, so training begins at the pretrained model. At $d_1 = d_2 = 768, r = 8$: 12,288 numbers against 589,824, 2.1%.
- **The change a task needs has low rank: one base model, one small versioned adapter per task.**

Domain-Adaptive Pretraining: Same Objective, Different Corpus

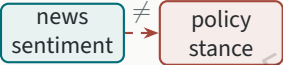


no new architecture at any step

- ▶ Continue the masking objective on text from the domain before adding a head (Gururangan et al. 2020).
- ▶ Why: *hawkish* in general text is a stance on war; in central-bank text it leans to raising rates.
- ▶ FinBERT (Araci 2019): BERT, then masking on TRC2-financial (46,143 Reuters articles, over 29M words), then a three-way head on the Financial PhraseBank (4,845 sentences).
- ▶ **Which corpus the model read is part of what the measure means.**

Example: A Domain Model's Labels Are Its Definition

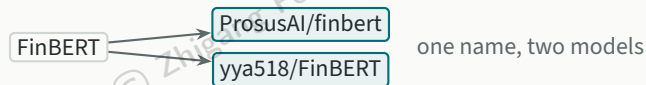
sentence, written for this slide	positive	negative	neutral
The Committee decided to raise the target range for the federal funds rate.	0.12	0.03	0.86
The Committee decided to lower the target range for the federal funds rate.	0.05	0.50	0.45
The Committee decided to maintain the target range for the federal funds rate.	0.06	0.03	0.91
The labor market has remained strong, and job gains have been solid.	0.95	0.03	0.02
Economic growth has slowed, and business investment has weakened.	0.01	0.97	0.02



- ProsusAI/finbert, recorded offline. A hike reads neutral (0.86), a cut negative (0.50): news for markets, not policy stance.
- **A classifier measures what its training labels meant: read the label set first.**

Two FinBERTs, and a Model With Two Classifiers

identifier	its paper	what it is
ProsusAI/finbert	Araci (2019)	financial news, then a three-way head on the Financial PhraseBank
yya518/FinBERT	Yang, UY & Huang (2020)	a different model of the same name, on financial communications
CentralBankRoBERTa	Pfeifer & Marohl (2023)	two classifiers: five macroeconomic agents, and a sign for each



- **Load a model by its identifier, cite the paper with it, and read its label set.**

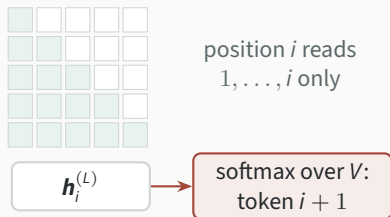
Session 5.3

The Decoder Route, and the Price of Scale

GPT: one next-token distribution, decoding, prompting, and what scale buys



Pretraining a Decoder: Predict the Next Token



- ▶ Same blocks with the causal mask: each row sees only its past, so it can predict what follows.
- ▶ $P_{\theta}(w_{i+1} \mid w_{1:i}) = \text{softmax}(\mathbf{W}_{\text{out}} \mathbf{h}_i^{(L)})$; **causal language modelling** maximises $\sum_i \log P_{\theta}(w_i \mid w_{<i})$.
- ▶ One nine-token sentence gives 8 predictions (The→central, The central→bank, ..., ... rising→inflation); masking 15% of it gives about 1.35.
- ▶ **A denser training signal, and only the left context: that trade separates GPT from BERT.**

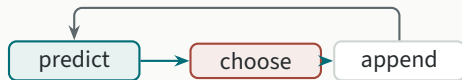
Example: A Next-Token Distribution, Recorded

next character	P	score
m	0.532	6.34
r	0.084	4.49
k	0.072	4.35
t	0.061	4.18
i	0.049	3.95
a	0.038	3.69

The Committee decided to_ **m**?

- ▶ **Machine B:** Session 1's text generator, trained on the 245 FOMC statements; 3 layers, $d = 128$, 631,424 parameters. Its tokens are 78 characters.
- ▶ After "The Committee decided to", it puts 0.53 on m , for *maintain*; then r (raise, reduce) and k (keep).
- ▶ Taking the top character each time: "... to maintain the target range for the federal funds rate at 0 to".
- ▶ **The model returns a distribution; a decoding rule turns it into text. Lab C4 runs it.**

Decoding: From a Distribution to Text



next	$\tau = 0.5$	$\tau = 1$	$\tau = 2$
m	0.93	0.53	0.19
r	0.02	0.08	0.07
k	0.02	0.07	0.07
t	0.01	0.06	0.06

- ▶ Generation loops: predict, choose one token, append it, predict again.
- ▶ **Greedy**: the top token. **Temperature**: $p_{\tau}(w) \propto \exp(s_w/\tau)$ on the scores s_w ; $\tau \rightarrow 0$ is greedy, $\tau > 1$ flattens. Top- k and top- p draw only from the best k tokens, or the smallest set with probability p .
- ▶ On Machine B's own scores (table): sampled at $\tau = 2$ it writes "neiqvelopments, both polifle advi2\$105. Biral 2 1.S5 fillp4".
- ▶ **For measurement set $\tau = 0$; report τ , the seed, the version and an input hash.**

In-Context Learning: The Task Goes in the Prompt

Label the monetary-policy stance of the Federal Open Market Committee statement below. Answer with exactly one word: HAWKISH if it leans toward tighter policy, DOVISH if it leans toward easier policy, NEUTRAL if it leans neither way. Output the single word and nothing else.

STATEMENT: [the statement of 1995-02-01: "The Federal Reserve Board today approved an increase in the discount rate from 4 3/4 percent to 5 1/4 percent, effective immediately." ...]

continuation

HAWKISH

recorded:
claude-sonnet-5,
2.1.278 (Claude
Code), 2026-09-20

- ▶ **Zero-shot:** an instruction and the item, as here. **Few-shot:** add worked examples first, e.g. "... decided to raise the target range. HAWKISH" (written for this slide).
- ▶ No parameter changes: the label is the model's continuation of the prompt.
- ▶ **The prompt is the estimator's specification: freeze it and publish it.**

In-Context Labels Carry No Internal Error Measure

estimation
fit to labels, then residuals $y_i - \hat{y}_i$



in context
an instruction and an item, one pass

nothing is fitted, so nothing inside reports an error

- ▶ Estimation fits parameters to labelled data and leaves regression residuals $y_i - \hat{y}_i$ to inspect.
- ▶ In-context labelling fits nothing: the model returns HAWKISH and no quantity says how likely it is to be wrong.
- ▶ Machine B's fluent statements had the same property: no target to subtract them from.
- ▶ **So the error is measured outside, on a hand-coded sample. Deck 5.4 builds it.**

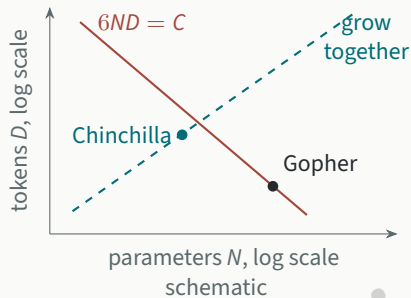
Two Paragraphs, Two Families

A, a policy statement: To counter rising inflation pressure, the central bank decided to raise the policy rate by 25 basis points. The Committee judges that maintaining a restrictive stance is appropriate until inflation returns to its target.

B, an annual report: During the reporting period, the company's financing costs increased as market interest rates rose. Management expects tighter credit conditions to put some pressure on next year's capital expenditure.

- ▶ **BERT**: pool each into $\mathbf{z}_A, \mathbf{z}_B$: cosine 0.72, against 0.46 with unrelated text; with labels, a stance head on $\mathbf{z}$.
- ▶ **GPT** (illustrative): "... the central bank decided to" continues *raise*; asked A's stance, "hawkish"; asked B's risk type, {risk type: rising financing costs}.
- ▶ *Rate* is the policy rate in A, a market rate in B; both read that from context.
- ▶ **BERT returns the same number every time; GPT moves with the prompt, decoding and version.**

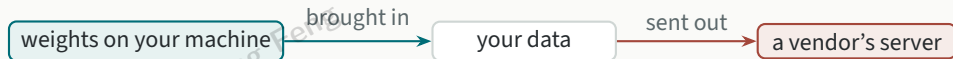
Scaling Laws, and What Chinchilla Corrected



- ▶ Held-out loss falls as a power law, $L(N) = (N_c/N)^{\alpha_N}$ with $\alpha_N \approx 0.076$: a straight line in logs over seven orders of magnitude (Kaplan et al. 2020).
- ▶ Hoffmann et al. (2022) fit $L(N, D) = E + A/N^{0.34} + B/D^{0.28}$ and minimise it subject to compute $C \approx 6ND$: double the parameters, double the tokens.
- ▶ Chinchilla, 70B parameters on 1.4T tokens (20 per parameter, $C = 5.9 \times 10^{23}$), beats Gopher, 280B on 300B ($1.1; 5.0 \times 10^{23}$).
- ▶ **A production function: parameters and tokens are inputs, $6ND$ is the isocost.**

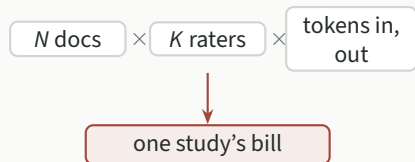
Open Weights, and Where the Data Is Allowed to Go

route	what you can archive	what it costs
a commercial interface	a version string, a call date and the sampling parameters	the text leaves the building, and the version moves underneath you
open weights, run locally	the weights themselves, beside the code and the prompt	your own hardware, and a step behind the frontier



- ▶ **Open weights:** the parameter files are published, so the model runs on your own hardware (LLaMA, Qwen, DeepSeek, GLM).
- ▶ **For confidential text the law decides, not accuracy: the model comes to the data.**

A Cost Formula, Not a Price List



per call it is nothing; per study it is a budget line

- ▶ $C = N K (t_{\text{in}} p_{\text{in}} + t_{\text{out}} p_{\text{out}})$: N documents, K raters or repeats, t tokens in and out per call, p the price per token on the day.
- ▶ In tokens: the lab's 245 statements average 425 tokens; with $K = 3$ and 5 tokens out, one pass reads and writes 316,050 tokens.
- ▶ Deck 5.5's replication of Yin, Vu & Persico (2026) runs 18,797 tasks with three raters.
- ▶ **Prices move every quarter: budget the study, and archive the day's quotation with the outputs.**

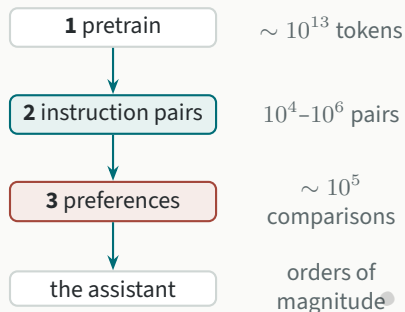
Session 5.3

Alignment

From a model that continues text to an assistant that answers



Why a Pretrained Decoder Is Not an Assistant



- ▶ Pretraining teaches continuation, not answers. Asked “How do I compute robust standard errors for a panel regression in Python?”, a base model may go on “Or: how do I do it in Stata...” (illustrative).
- ▶ Alignment adds two stages on far less data: fine-tuning on instruction–answer pairs, then learning from preferences.
- ▶ InstructGPT used about 13k prompts to fine-tune and 33k for its reward model (Ouyang et al. 2022).
- ▶ **Small stages that set how the model answers everything, your labels included.**

Stage 1: Supervised Fine-Tuning

x : Classify the stance of “The Committee decided to raise the target range.” One word.

y : Hawk ish .

each answer token scored given x and the earlier answer tokens (an example written for this slide)

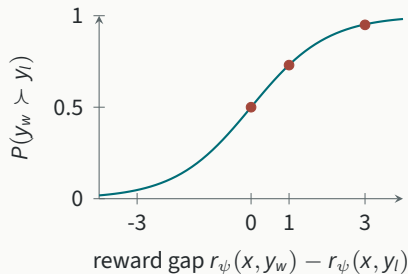
- ▶ Data: pairs (x, y) , an instruction x and a written answer $y = (y_1, \dots, y_m)$.
- ▶ Loss on the answer tokens only:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y)} \sum_{j=1}^m \log P_{\theta}(y_j \mid x, y_{<j}),$$

the next-token loss, on instruction data.

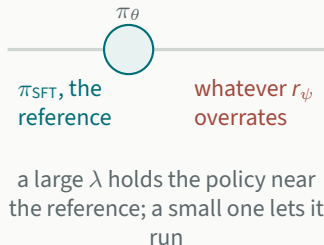
- ▶ It fixes form: format, when to decline, how far to reason.
- ▶ **Each instruction carries one answer, so the model never learns why another would be worse.**

Stage 2a: A Reward Model From Comparisons



- ▶ Annotators see two replies to one instruction x and pick one: y_w preferred to y_l .
- ▶ Bradley-Terry:
 $P(y_w \succ y_l | x) = \sigma(r_\psi(x, y_w) - r_\psi(x, y_l))$; r_ψ is the pretrained network with a one-number head, fit by minimising $-\sum \log \sigma(r_\psi(x, y_w) - r_\psi(x, y_l))$.
- ▶ Worked: reward gaps of 0, 1 and 3 give 0.50, 0.73 and 0.95.
- ▶ **A binary logit: utility r_ψ plus type-I extreme-value noise (Luce-McFadden). Deck 5.4 fits the same model to coded survey answers.**

Stage 2b: Optimise Against the Reward, on a Leash

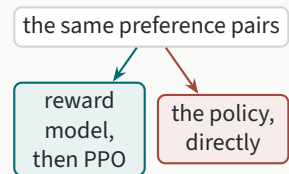


- ▶ The **policy** $\pi_\theta(y | x)$: the model's probability of reply y to prompt x .

$$\max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [r_\psi(x, y)] - \lambda \mathbb{E}_{y \sim \pi_\theta} \log \frac{\pi_\theta(y | x)}{\pi_{\text{SFT}}(y | x)}$$

- ▶ The second term is the KL divergence from the fine-tuned model; without it the policy finds replies r_ψ overrates: **reward hacking**.
- ▶ PPO (Deck 3.5) solves it, clipping the ratio $\pi_\theta / \pi_{\text{old}}$ to $[1 - \epsilon, 1 + \epsilon]$ so no step moves the policy far.
- ▶ **Ridge for a policy:** $\|y - X\beta\|^2 + \lambda \|\beta - \beta_0\|^2$ shrinks to β_0 ; here π_{SFT} is β_0 .

Direct Preference Optimisation: The Same Pairs, No Reward Model



two routes, one object: what the annotators preferred

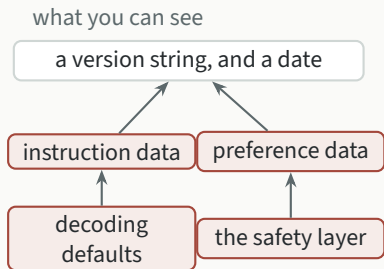
- ▶ The least problem's optimum:
 $\pi^*(y | x) = \pi_{\text{SFT}}(y | x) e^{r(x,y)/\lambda} / Z(x)$, $Z(x)$ the normaliser.
- ▶ Invert it, $r = \lambda \log(\pi^* / \pi_{\text{SFT}}) + \lambda \log Z(x)$, and put it in the Bradley-Terry gap, where $Z(x)$ cancels:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E} \log \sigma \left(\lambda \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{SFT}}(y_w|x)} - \lambda \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{SFT}}(y_l|x)} \right)$$

(Rafailov et al. 2023, who write β for λ).

- ▶ **Raise the preferred reply relative to the reference, lower the rejected one: the same object, no reward model. Llama 3 (Grattafiori et al. 2024) and Qwen2 (Yang et al. 2024) use it.**

The Version Is the Instrument



what moved underneath it: private, rolling, and not announced

- ▶ Instruction and preference data are the vendor's, private and rolling: from outside you see a version string, such as `gpt-4-0613` OR `gpt-4o-2024-08-06`.
- ▶ Two studies that both “used GPT-4” may have used different instruments, and neither can tell.
- ▶ Deck 5.5 scores one frozen rubric with several raters and shows how far the readings move.
- ▶ **An aligned model used for scoring is a time-varying instrument: the version is part of the measure's definition.**

Session 5.3

Five Limits

What the architecture implies, and what no amount of scale removes

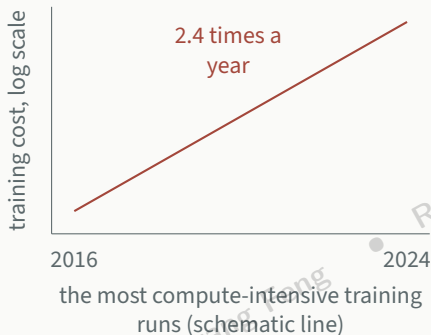


Five Limits, Five Architectural Roots

limit	root in the architecture	what it does to a design
the cutoff	knowledge sits in the weights	put the document in the prompt
hallucination	the objective rewards form	check every citation and number
the window	attention costs $O(n^2)$	blocks sever cross-references
lost in the middle	position bias in attention	position is part of the error
prompt sensitivity	the prompt is the prior	freeze it, and vary the order

- ▶ None is a defect awaiting a patch; each follows from how the machine is built.
- ▶ **A larger model moves all five, and removes none of them.**

Limit One: The Cutoff



- ▶ Everything the model knows sits in its weights, fixed on a date the vendor publishes. Later events are invisible to it.
- ▶ Updating knowledge means retraining, and the cost of the largest training runs has grown 2.4 times a year since 2016 (Cottier et al. 2025).
- ▶ **So the repair is not retraining: put the passage in the prompt. Session 6 makes that a system.**

Limit Two: Fluency Is Not Truth

what is maximised:

$$\log P_{\theta}(w_i \mid w_{<i})$$

a property of the *language*

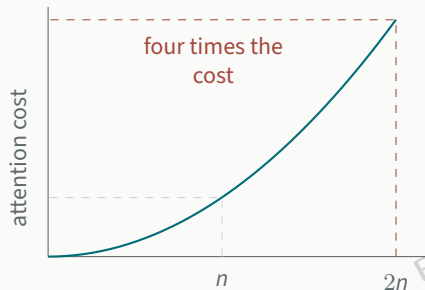
what you wanted:

a property of the *world*

where the training text was thin, the
form is filled with invention

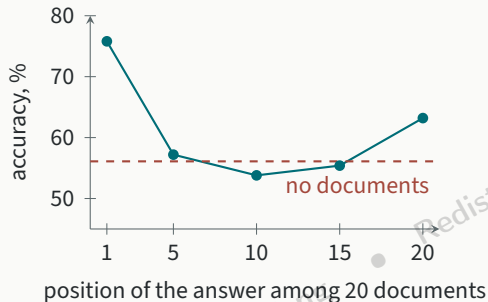
- ▶ **Hallucination:** fluent output that is false — invented citations, plausible numbers, laws that do not exist.
- ▶ The objective scores how likely the continuation is, never whether it is true.
- ▶ A citation's form, "Author (Year), *Title*, Journal 34(2): 101–120", is predictable token by token: every token can be likely while the whole is false.
- ▶ **Every citation, number and quotation is a draft until it is checked outside the model.**

Limit Three: The Window Is Quadratic



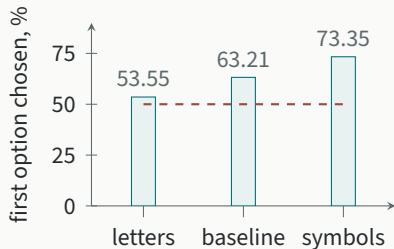
- ▶ Attention scores every position against every other: cost $O(n^2)$. A median FOMC statement of 410 tokens already needs 168,100 scores per head and layer.
- ▶ The window is a hard bound: one annual report fits, a decade of transcripts does not.
- ▶ Cutting a document into blocks severs the reference from the management discussion back to the risk factors.
- ▶ **Sparse attention and state-space models (Gu & Dao 2023) cut the cost and give up exact global attention.**

Limit Four: Lost in the Middle



- ▶ Move the document that holds the answer through 20 retrieved documents: accuracy traces a U, best at the ends, worst in the middle (Liu et al. 2024; GPT-3.5-Turbo, their Table 6).
- ▶ In the middle it does worse than with no documents at all: more context is not more information.
- ▶ Base models show the same curve.
- ▶ **Non-classical error: in long and short filings the noise differs, by position.**

Limit Five: The Prompt Is the Prior



- ▶ Where order should not matter, GPT-4 picked the first option 63.21 per cent of the time; 53.55 with letter labels, 73.35 with symbols (Brucks & Toubia 2025).
- ▶ Saying the order was randomised does not repair it: 63.82 against 65.00 per cent.
- ▶ The same pull towards the ends as the long-context study, on a list of labels.
- ▶ **Freeze the prompt, version it, publish it, vary the order — Deck 5.4 turns all five limits into a protocol.**

Session 5.4

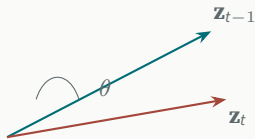
The LLM as a Measurement Instrument

What it can label, what the labels are worth, and how to correct what they get wrong



Zhigang Feng

Cosine Similarity as an Economic Variable



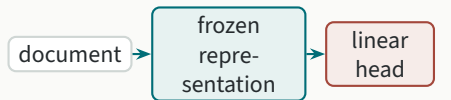
$\text{sim} = \cos \theta$: same direction
1, orthogonal 0, opposed -1

$\delta_t = 1 - \text{sim}(\mathbf{z}_{t-1}, \mathbf{z}_t)$
how far this statement moved

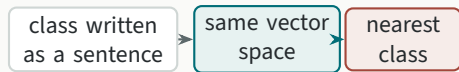
- ▶ $\text{sim}(i, j) = \mathbf{z}_i^\top \mathbf{z}_j / (\|\mathbf{z}_i\| \|\mathbf{z}_j\|)$ is invariant to length, and length tracks firm size.
- ▶ **Cosine distance** over consecutive statements is continuous where a hawk-dove label is binary.
- ▶ To call δ_t a shock, clear two layers: what markets priced, and whether the bank merely responded to conditions.
- ▶ **The geometry gives a number.**
Whether it is a shock is a claim about the design.

Source: Miao & Feng, *AI for Economists*, ch. 13, “语义相似度计算”; Acosta & Meade (2015).

Document Classification, With Labels and Without



needs a few hundred labels



needs none: $\hat{c}_i = \arg \max_k \text{sim}(\mathbf{z}_i, \mathbf{z}_{c_k})$

- ▶ **Supervised:** freeze the representation, fit a linear head on a few hundred labels. The labels are the binding constraint.
- ▶ **Zero-shot:** embed a class description with the document; take the nearest class.
- ▶ Raw encoder vectors are anisotropic: sentences sharing frequent words look close.
- ▶ **Needing no labels is paid for in validation: the class wording moves the result.**

Source: Miao & Feng, *AI for Economists*, ch. 13; Reimers & Gurevych (2019).

The Standard These Measures Are Held To

measure	study	text operation	how it was validated
corporate culture	Li, Mai, Shen & Yan (2021)	a dictionary grown from seed words	later performance; enforcement actions
macro nowcast	Bybee, Kelly & Manela (2023)	topic shares as predictors	added forecast power for output
political risk	Hassan & Hollander (2019)	counts of political word pairs	lobbying spending
communication change	Acosta & Meade (2015)	cosine distance between statements	policy turning points; descriptive

- **The last column is what makes any of them a measure — and every row still yields a predictive proxy, not an identified effect.**

Source: Miao & Feng, *AI for Economists*, ch. 13, table of text measures in economics.

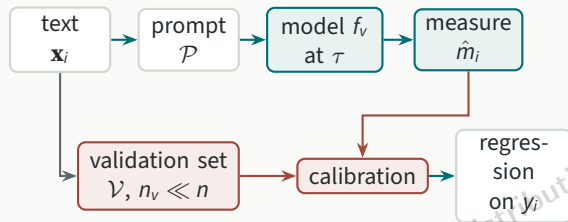
Session 5.4

The Instrument

A prompt, a model version and a temperature: where a text measure comes from now



The Measurement Chain, End to End



- ▶ Every box is a decision that has to be reported: corpus, prompt text, model version, temperature, and how $\mathcal{V}$ was drawn.
- ▶ $\mathcal{V}$ is the only link that puts $\hat{m}_i$ back on the scale of the construct.
- ▶ **Everything to the right of the measure is econometrics, not language modelling.**

Source: Miao & Feng, *AI for Economists*, ch. 14, figure of the measurement chain.

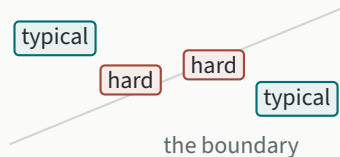
Fine-Tune an Encoder, or Prompt a Model

critterion	fine-tuned encoder	prompted model
labels	hundreds in hand, or affordable	none, or costlier than the answer
task definition	settled; the manual is written	still being revised
sample size	hundreds of thousands	thousands to tens of thousands
reproducibility	frozen weights, reruns to the digit	version disclosed, robustness re-checked
what it leans on	dense domain vocabulary	common-sense and pragmatic inference

- **Cost structure decides this, not accuracy: fine-tuning owns an instrument, prompting rents one. Or prompt a few hundred cases to settle the definitions, then fine-tune on them.**

Source: Miao & Feng, *AI for Economists*, ch. 14, table comparing encoder fine-tuning with prompting.

Zero- and Few-Shot Labelling, and Which Examples to Show



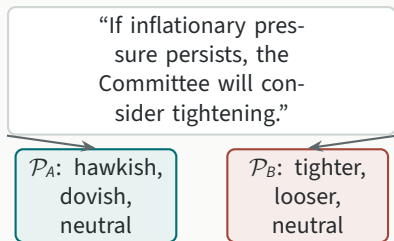
a second obvious example teaches the centre again; a hard one moves the boundary

θ fixed, $\mathcal{P}$ carries the task

- ▶ One equation for the whole section: $\hat{m}_i = \arg \max_c p(c \mid \mathbf{x}_i, \mathcal{P}; \theta)$, with θ fixed and the prompt doing every bit of task adaptation.
- ▶ **Zero-shot** removes the fixed cost of labelling; **few-shot** buys calibration with a few demonstrations.
- ▶ **Pick demonstrations at the boundary, not at the centre:** consistency across a sample is decided by the hard cases.

Source: Miao & Feng, *AI for Economists*, ch. 14, “零样本与少样本标注”; Gilardi et al. (2023).

One Statement, Two Wordings



a stance

a leaning

the drift is systematic: inter-coder reliability cannot see it

claude-sonnet-5, Claude Code 2.1.278 · 2026-09-20 ·
prompts 846fe144, 3307f8f3, a47b80db

- ▶ The sentence is a conditional: nothing is announced.
- ▶ $\mathcal{P}_A$ folds “hawkish” into “announces tightening”; $\mathcal{P}_B$ leaves room for a leaning. On an unconditional sentence the two agree.
- ▶ This session’s own stance run, 4 prompt variants over 48 statements: κ 0.746 rewoded, 0.843 reordered, 0.717 on an identical repeat.
- ▶ **Write conditionals into the class definition and test on boundary cases.**

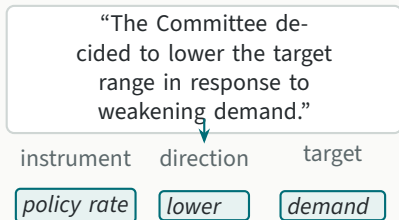
Source: Miao & Feng, *AI for Economists*, ch. 14, example box on prompt perturbation.

Sentiment: Where the Advantage Actually Comes From

- ▶ A dictionary reads a document as an unordered set of words: negation, conditionals and irony sit outside the model rather than being handled badly by it.
- ▶ **Reach**. Nineteenth-century newspapers are dense with scanning error, where keyword matching is brittle and a semantic judgement is not. The gain is which archives become usable at all.
- ▶ **Excess predictive power**. Daily firm news scored by a model and sorted into a long-short portfolio keeps a positive return spread after costs — and tell the model to judge words literally instead, and that power disappears.
- ▶ **Structured sentiment**. Split macro news into drivers — inflation worry, supply-chain strain, weak demand — and the nowcast improves where one positive-negative score does not.
- ▶ **The advantage is context, which is also where the error is hardest to diagnose from word counts.**

Source: Loughran & McDonald (2011); Lopez-Lira & Tang (2023); Hartley (2025); BIS (2025).

Structured Extraction: Fields, Not Labels



one record per document,
with a fixed set of fields —
an empty slot is information

- ▶ The output is a record, not a class: acquirer, target, value, date.
- ▶ Prompted extraction beats few-shot baselines on spans, tokens and relations in clinical text — jargon-dense and inconsistently named, as economic text is.
- ▶ A narrow templated task still favours a small fine-tuned model.
- ▶ **Session 6 turns this record into a typed tool call, where the schema is enforced.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “结构化抽取”; Agrawal et al. (2022).

Session 5.4

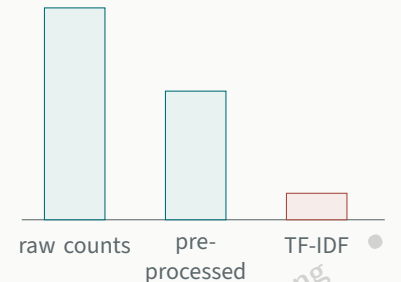
Measures That Were Estimated

Similarity, topic shares and labels are all output of a first stage



The Same Two Statements Under Three Weightings

similarity to the previous statement

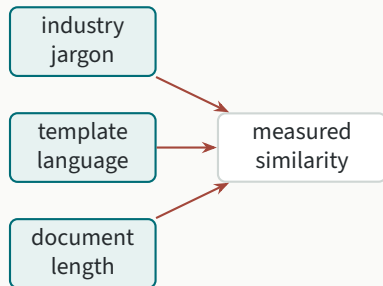


one pair of documents, three answers

- ▶ The December 2008 statement — the meeting that took the funds-rate target to its floor — reads as a light edit on raw counts and as a near-complete rewrite under TF-IDF.
- ▶ Raw counts are dominated by shared boilerplate; TF-IDF gives its weight to what only this statement says.
- ▶ **Deck 5.1's pre-processing choices do not wash out downstream: they set the size of the variable you are about to regress.**

Source: Acosta & Meade (2015), consecutive statements under three weightings; Miao & Feng, ch. 13.

Three Confounds Inside a Similarity



all three push it up,
none of them is strategy

- ▶ **Industry:** one industry shares a vocabulary, so similarity is high before any strategy is compared.
- ▶ **Template:** mandated disclosure language repeats verbatim, with nothing behind it.
- ▶ **Length:** a longer document has a steadier average vector and looks closer to everything.
- ▶ **Deck 2.3's leakage frame is the template confound:** template sentences scored 0.97 on a random split and 0.50 — chance — with phrases held out.

Source: Miao & Feng, *AI for Economists*, ch. 13, “验证挑战：三类混淆”; Session 2 lab notebook, appendix.

What to Do About Them

permutation

shuffle labels
inside an
industry

is it above
the baseline?

residual

subtract the
industry mean

remove the
shared part

external

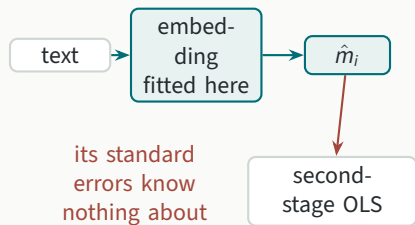
tie it to an
observable
outcome

is it a measure
of anything?

- ▶ **Permutation test:** shuffle firm labels within an industry, recompute the similarity distribution, and ask whether the observed value clears that baseline.
- ▶ **Residual embedding:** subtract the industry mean vector before measuring.
- ▶ **External validation:** relate the similarity to merger incidence or joint research spending.
- ▶ **The first two remove a confound. Only the third says the measure is of anything.**

Source: Miao & Feng, *AI for Economists*, ch. 13; Gentzkow, Kelly & Taddy (2019).

The Generated-Regressor Problem



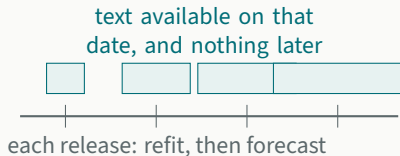
its standard errors know nothing about the first stage

the bootstrap resamples the whole chain, not the last link

- ▶ A similarity, a topic share, a model score: each is an estimate carrying error from the model and the pooling rule.
- ▶ Treat it as observed and the second-stage standard errors are too small.
- ▶ Worst where the first stage is small: an embedding fitted on a few hundred statements.
- ▶ **Resample documents, re-embed, re-estimate, and report that interval.**

Source: Miao & Feng, *AI for Economists*, ch. 13, “生成回归量问题”.

Re-Estimating at Each Release Is a Design, Not a Detail

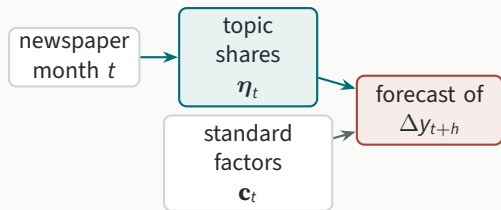


one embedding fitted on everything leaks the future into the representation

- ▶ The euro-area exercise refits its word vectors at every press conference, from the text available that day, and forecasts core inflation from them.
- ▶ Fitting it once on the full sample leaks the future into the representation, not just the regression.
- ▶ Session 2's leakage rule reaches the first stage: the vectors belong to the training window.
- ▶ **It costs compute and buys a claim the forecast could have been made.**

Source: Araujo, Bokan, Comazzi & Lenza, real-time word vectors for inflation forecasting.

Nowcasting: Text Factors Enter Like Any Other Factor



η_t came out of a first-stage fit: the interval has to be bootstrapped through it

- ▶ Quarterly output is published about a quarter late; a nowcast has to speak before the release.
- ▶ Monthly topic shares enter the usual predictive regression beside surveys and lagged output:

$$\Delta y_{t+h} = \alpha + \beta^\top \eta_t + \gamma^\top \mathbf{c}_t + \varepsilon_{t+h}.$$

- ▶ **Text still adds forecast power with the standard factors controlled — but bootstrap text, topics and regression together.**

Source: Bybee, Kelly & Manela (2023); Miao & Feng, *AI for Economists*, ch. 13.

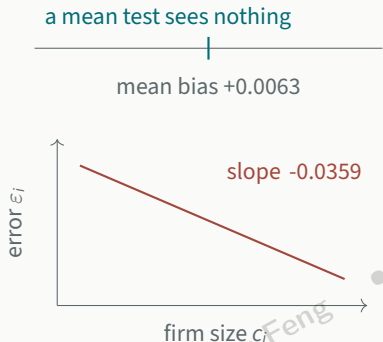
Session 5.4

The Error, and What to Do About It

A validation set, a calibration equation, and a budget for hand coding



The Error Is Not the Classical Kind



- ▶ $\hat{m}_i = m_i^* + \varepsilon_i$, and ε_i moves with m_i^* (models avoid extreme judgements) and with the covariates.
- ▶ On the book's simulated reports the measure correlates 0.851 with the truth at mean bias +0.0063: a mean test passes it.
- ▶ The error on firm size has slope -0.0359, standard error 0.0019: large firms are understated.
- ▶ **What matters is the covariance of the error with the regressors, not its mean.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “测量误差的结构”; Ludwig & Mullainathan (2025).

On the Right-Hand Side, or on the Left

$$y_i = \alpha + \beta \hat{m}_i + \gamma^\top \mathbf{c}_i + u_i$$

biased; the bias leaks into γ as well

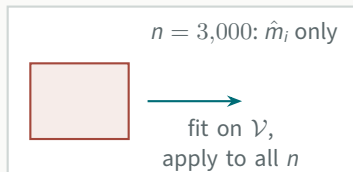
$$\hat{m}_i = \alpha + \gamma^\top \mathbf{c}_i + u_i$$

consistent if $\mathbb{E}[\varepsilon_i \mid \mathbf{c}_i] = 0$;
only precision is lost

- ▶ As a **regressor**, $\hat{m}_i$ brings attenuation plus a non-classical part whose direction cannot be signed in advance.
- ▶ As an **outcome** — what makes a firm write this way? — least squares stays consistent as long as the error has conditional mean zero.
- ▶ **Opposite diagnostics: on the right, test the error against the regressors; on the left, test its conditional mean. Carrying one check across is the common mistake.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “测量误差的结构”.

Correcting It With a Human Validation Set



$\mathcal{V}: n_v = 200$,
both m_i^* and $\hat{m}_i$

- ▶ Hand-code a validation set $\mathcal{V}$ of size $n_v \ll n$ carrying both the human m_i^* and the model's $\hat{m}_i$.
- ▶ Step one, on $\mathcal{V}$: $\hat{m}_i^* = \hat{a}_0 + \hat{a}_1 \hat{m}_i + \hat{\mathbf{a}}_2^\top \mathbf{c}_i$.
Step two, on all n : regress y_i on $\hat{m}_i^*$ and $\mathbf{c}_i$.
- ▶ $\mathbf{c}_i$ has to be in the calibration equation — it is what absorbs the systematic link between the error and the covariates.
- ▶ **Do not use the second step's textbook standard errors: the calibration is itself estimated, so bootstrap both samples.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “用人工验证集做偏误修正”.

What the Correction Assumes

random draw

$\mathcal{V}$ from the full sample

coding the easy ones first breaks it

extrapolation

the fit holds outside $\mathcal{V}$

rare wording may not be in $\mathcal{V}$

exclusion

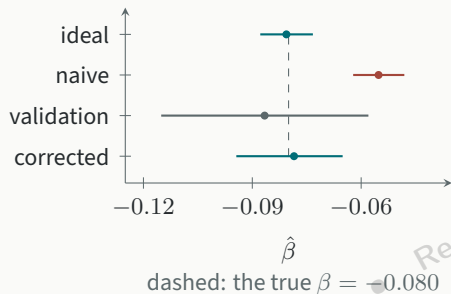
the residual of $\hat{m}$ is inert

what is left must not reach y_i

- **One:** $\mathcal{V}$ is a random draw. Hand-coding whatever is quickest to read breaks it.
- **Two:** the calibration extrapolates beyond $\mathcal{V}$. If the model fails only on unusual wording, $\mathcal{V}$ may contain none of it.
- **Three:** given $\hat{m}_i$ and $\mathbf{c}_i$, the relation between m_i^* and y_i no longer depends on what is left in $\hat{m}_i$.
- **State all three. Each is a claim about how the labels were made, which only the researcher can answer.**

Source: Miao & Feng, *AI for Economists*, ch. 14, the three assumptions of the two-step estimator.

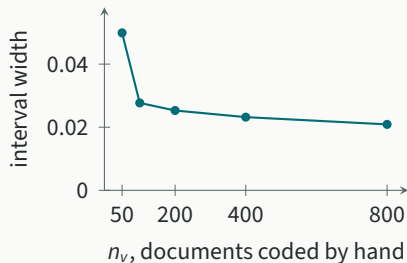
Four Estimators, and the Slope That Explains Them



- ▶ Ideal, on m_i^* : -0.0806. Naive, on $\hat{m}_i$: -0.0552, 31.0 per cent from the truth and shrunk toward zero.
- ▶ Calibration on $\mathcal{V}$: $0.1423 + 0.7031 \hat{m} + 0.0048 c$, and that slope below one *is* the attenuation.
- ▶ Validation set alone -0.0866 at width 0.0570; corrected -0.0785, 1.9 per cent out, at 0.0293 — 49 per cent narrower.
- ▶ **200 hand-coded documents buy back full-sample precision.**

Source: Miao & Feng, *AI for Economists*, ch. 14, worked example on simulated annual reports.

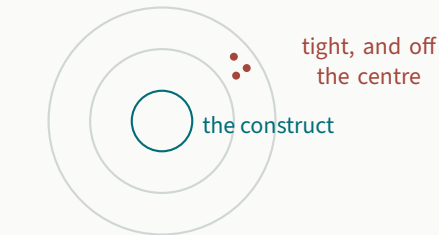
How Many Documents to Hand-Code



- ▶ Width falls in n_v at a decreasing rate: 49 per cent narrower from 50 to 200 labels, a further 17 per cent from 200 to 800.
- ▶ At small n_v the calibration's own error dominates; past a few hundred the full-sample residual variance binds.
- ▶ Hold the two 49 per cents apart: the last page compared two *estimators* at $n_v = 200$; this compares two *validation sizes* for one estimator.
- ▶ **Buy enough labels to kill the bias, then stop enlarging the set.**

Source: Miao & Feng, *AI for Economists*, ch. 14, design calculation for the validation set.

Consistency Is Not Validity

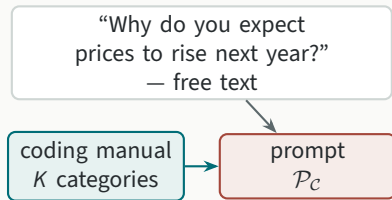


κ measures the spread of the darts, never the distance to the middle

- ▶ κ compares two coding processes; it is silent on whether either measures the construct.
- ▶ A model can be perfectly consistent about high and low commitment and still be unrelated to green investment.
- ▶ If the manual defines the construct badly, both coders agree on the wrong definition.
- ▶ **Validity is argued from outside: later emissions, capital expenditure, compliance.**

Source: Miao & Feng, *AI for Economists*, ch. 14, on reliability and construct validity.

Coding Open Survey Answers



the prompt is the manual's implementation: where it is vague the model fills the gap consistently

- ▶ Narrative heterogeneity is the interest: do different attributions save and spend differently?
- ▶ Categories come from the manual *before* the data, or they become part of the result.
- ▶ The prompt must reproduce the operational definitions and the exclusion clauses.
- ▶ **Four rules: temperature zero; a real “cannot be coded” category; reasons before the label; the prompt archived.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “开放式问题的自动编码”; Haaland et al. (2025).

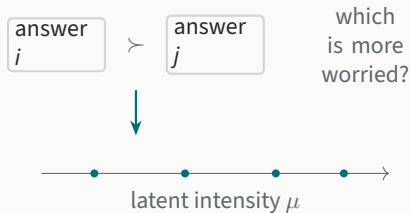
Report κ by Class, Never Overall Alone

category	share	recall	κ	at the 0.75 threshold
supply-chain narrative	0.413	0.968	0.891	passes
monetary narrative	0.387	0.974	0.876	passes
opportunistic-pricing narrative	0.120	0.528	0.634	fails; act
cannot be coded	0.080	0.750	0.825	passes
overall	1.000	0.900	0.844	reads as excellent

- ▶ A model is low-variance and possibly biased, so a repeat call adds nothing: audit 10 per cent of the 3,000 answers blind against a human gold standard.
- ▶ **Half of the low-frequency narrative is misfiled and the overall row hides it. Merging it lifts κ to 0.881 — by abolishing the distinction.**

Source: Miao & Feng, *AI for Economists*, ch. 14, per-class audit on simulated survey answers.

Intensity by Pairwise Comparison



the origin is not identified: only differences mean anything

- ▶ When the target is intensity, a direct score out of a hundred has no common scale; a pairwise judgement does.
- ▶ $\Pr(\mathbf{x}_i \succ \mathbf{x}_j) = e^{\mu_i} / (e^{\mu_i} + e^{\mu_j})$:
Bradley–Terry, with μ by maximum likelihood and a randomised block design.
- ▶ Deck 5.3 fitted this model to human preferences over model outputs; here a model ranks human answers.
- ▶ **μ is identified only up to a shift: standardise it or use the ranks.**

Source: Bradley & Terry (1952); Miao & Feng, *AI for Economists*, ch. 14, “连续强度的编码”.

The Validation Ladder

- ▶ **Pre-alignment:** hand-code one to two hundred answers and iterate the prompt against them — then retire them. They tuned the instrument, so they cannot grade it.
- ▶ **Gold-standard audit:** once the full sample is coded, blind-recode at least 10 per cent of it and compute Cohen's κ against the human labels.
- ▶ **Per-class report:** κ and the confusion matrix category by category. Below the threshold the fallback is hand coding or a coarser category, and the cost falls on the research question rather than on the statistic.
- ▶ One κ , three adjectives: the audit's 0.844 is excellent on the bands of Fleiss (1981), almost perfect on the 0.81–1.00 band of Landis & Koch (1977), and strong on McHugh (2012). The 0.75 threshold is Fleiss's; Landis & Koch has no such cut.
- ▶ **Name the scheme whenever a κ takes an adjective, and finish the ladder: the last rung is external validity, which none of the rungs below can supply.**

Source: Fleiss (1981); Landis & Koch (1977); McHugh (2012); Miao & Feng, *AI for Economists*, ch. 14.

Session 5.5

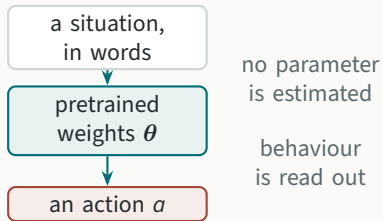
Simulated Agents, Prompts as Design, and Reproducible LLM Research

What a model can stand in for, why the prompt is a confounder, and what has to be reported



Zhigang Feng

Homo Silicus: A Subject You Can Query

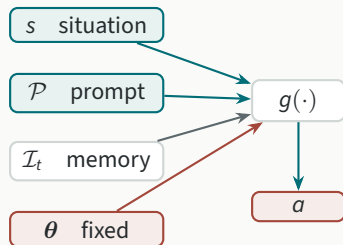


the weights hold the
behaviour text *records*

- ▶ **Homo silicus**: the model as an implicit computational model of human behaviour, queried in prose rather than estimated.
- ▶ It took fairness, reciprocity and loss aversion from human writing, so it answers a classic game in the human direction.
- ▶ **But nothing separates generalisation from recall of the published experiments.**

Source: Horton (2023), homo silicus; Miao & Feng, *AI for Economists*, ch. 14, “LLM 作为模拟经济主体”.

The Agent's Policy: What You Move, and What You Cannot



two of the four are
prose you write

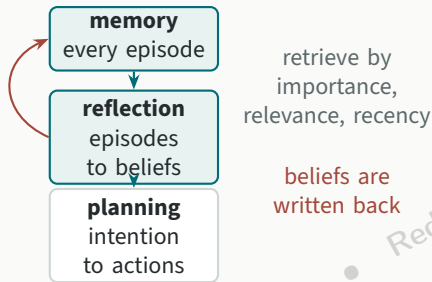
- ▶ Behaviour is a conditional distribution over actions:

$$a \sim g(\cdot \mid s, \mathcal{P}, \mathcal{I}_t; \theta).$$

- ▶ You move two pieces of prose, s and $\mathcal{P}$, not the parameters of a utility function.
- ▶ θ is fixed, so all behavioural heterogeneity enters through the role instruction.
- ▶ **The role instruction does the work a preference parameter used to do.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “生成式智能体模拟”.

Memory, Reflection, Planning: Where $\mathcal{I}_t$ Comes From



all three together are $\mathcal{I}_t$

- ▶ The base model is stateless. Whatever must persist across rounds lives in $\mathcal{I}_t$, outside the model.
- ▶ Retrieval scores memories on importance, relevance and recency; reflection turns episodes into beliefs.
- ▶ **How many memories, at what weights, how often reflection runs: design parameters. Undisclosed, two runs are not comparable.**

Source: Park, O'Brien, Cai, Morris, Liang & Bernstein (2023), generative agents.

Three Tiers of Evidence, and What Each One Licenses

established: pre-testing

emerging: a calibrated twin

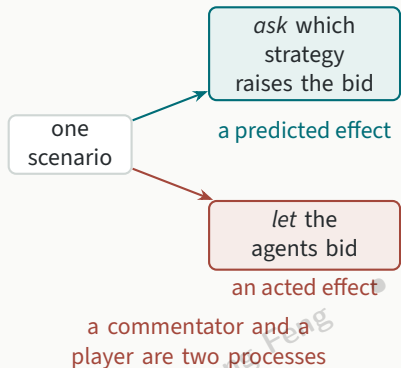
insufficient: replacing subjects

a conclusion may rest
above the line, not below it

- ▶ **Established:** screening treatment variants. It touches the design, never the inference.
- ▶ **Emerging:** a twin calibrated on a long interview, reproducing one person's answers on particular scales.
- ▶ **Insufficient:** silicon subjects as the main source for a causal claim.
- ▶ **The line is conclusion strength, not model power: the same simulation is tier one screening and tier three estimating.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “按证据成熟度划分应用场景”.

Three Threats to External Validity, and a Fourth Difference



- ▶ **Memorised replication:** human-like play in the classic games may be recall of published results. Test on a design the corpus never saw.
- ▶ **Corpus representativeness:** English and Western text dominate, and the substitution leaves no trace in the output.
- ▶ **Version instability:** “we used model X” is an incomplete method description.
- ▶ **A predicted effect is not an acted one: the unit of analysis is behaviour.**

Source: Manning, Zhu & Horton (2024), automated social science; Argyle et al., *Political Analysis* (2023).

How to Write the Sentence, and Whose Consent It Needs

“the model *has* a fairness preference”

a theoretical claim:
no one can check it

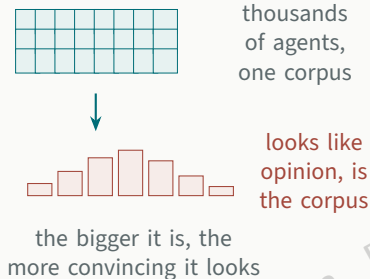
“under prompt $\mathcal{P}$ it produced outputs rejecting low offers”

an observable: the same prompt reproduces it

- ▶ Talking to it in prose is **functional anthropomorphism** and is fine. Claiming it *has* an internal state is an unsupported assertion.
- ▶ **Consent:** a twin built from one interview extends that person’s own data, so an identifiable simulation needs consent even with no human in the room.
- ▶ **“Analysis for this study” does not cover building a twin and querying it for years.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “伦理考量”.

Synthetic Consensus, and What a Pre-registration Would Hold



- ▶ **Synthetic consensus:** a large agent network's output can look like a distribution of opinion while being the corpus's content preferences.
- ▶ Keep a human check at every policy-relevant node, and verify against the population.
- ▶ **Pre-registration, unsettled:** is temperature the randomisation? Is the sample runs, agents or scenario pairs? Does a causal model written in advance replace a pre-analysis plan?

Source: Miao & Feng, *AI for Economists*, ch. 14, “合成共识”.

Session 5.5

The Model in Your Own Process

Where it changes the cost of the work, and where it must not touch the judgement



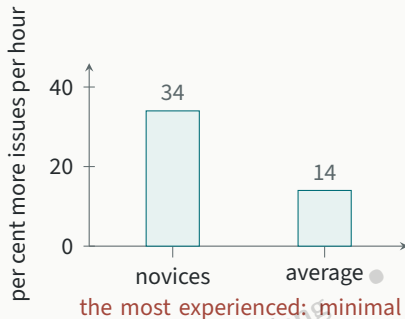
Five Task Modules, Ordered by What You Can Hand Over

module	what it covers	how far you can lean on it
literature	retrieval and comparison frames	Far: a wrong citation shows up on checking
data	formats, matching, cleaning	Far, with assertions: nothing raises
code	prototypes, translation, errors	Far: running it exposes most errors
writing	drafts, rewriting, captions	Far: the author owns every word
identification	question, strategy, causation	Not at all: institutional knowledge

- ▶ The first four are repetitive cognitive labour; the fifth is the academic contribution.
- ▶ **The red line: once a step's output decides the identification strategy, that step is yours.**

Source: Miao & Feng, *AI for Economists*, ch. 14, table of research tasks; Korinek, *JEL* (2023).

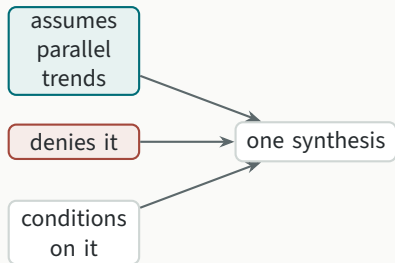
What the Productivity Evidence Actually Shows



- ▶ Writing tasks, 453 professionals randomly given access: time down 40 per cent, quality up 18 per cent, the spread narrower.
- ▶ A standardised server-programming task: 55.8 per cent faster.
- ▶ Customer support, 5,179 agents: 14 per cent more issues per hour, 34 per cent for novices.
- ▶ **Cognitive levelling: the gain sits at the inexperienced end.**

Source: Noy & Zhang, *Science* 381(6654):187–192 (2023); Peng et al. (2023); Brynjolfsson, Li & Raymond, *QJE* (2025).

The Synthesis Trap



the disagreement was
the point, and it is gone

- ▶ Past a few hundred papers the bottleneck moves from finding to integrating.
- ▶ The model smooths theoretical disagreement into an over-harmonious narrative. Alignment prefers uncontested statements, so a better prompt does not remove it.
- ▶ **Invert the task:** ask which papers disagree on the same identifying assumption and who is on each side.

Source: Miao & Feng, *AI for Economists*, ch. 14, “整合的陷阱”.

It Will Go Along with a Flawed Design



nothing is raised; agreement leaves no trace



the same model, turned on its own output

- ▶ **Silent errors:** a merge that drops rows on a case mismatch, a conversion the wrong way round, a window off by one. They propagate as bias.
- ▶ **Leakage:** pretraining holds the outcome, so the sentiment of a past date comes back contaminated by hindsight.
- ▶ Hand it an endogenous design and it writes working code rather than naming the problem.
- ▶ **Assertions after every step, and the model turned on its own output.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “数据与代码环节的两类静默风险”.

Session 5.5

The Prompt Is the Instrument

Why the wording is a design variable, and what this course measured on its own



The Prompt Is a Confounder, Not an Interface

prompt $\mathcal{P}_1$



same model,
same document

prompt $\mathcal{P}_2$



two instruments, not two phrasings

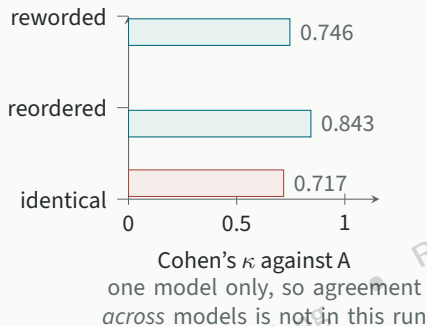
- ▶ Hold the model and the input fixed. The prompt still moves the output distribution:

$$\partial \mathbb{E}[\hat{m}_i \mid \mathbf{x}_i, f_v] / \partial \mathcal{P} \neq 0.$$

- ▶ A criterion, not a model. Where that derivative is non-zero the prompt belongs in the methods section beside the sample rule.
- ▶ **The prompt does the job the instruction sheet does in a laboratory experiment.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “提示选择作为混淆变量”.

This Course Measured Its Own Instrument

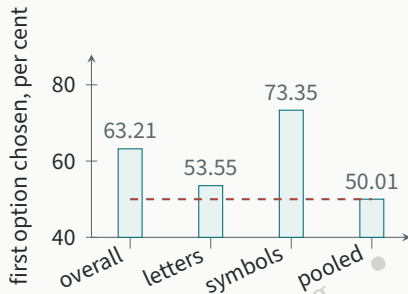


- ▶ One frozen prompt on 245 statements; three variants on a stratified 48: B reworted, C reordered, **D byte-identical to A.**
- ▶ Agreement with A: 83, 90 and 81 per cent. Not pinned: the harness's own system prompt, and the temperature.
- ▶ **The identical prompt agrees with itself least. Without D, B's gap reads as wording — and reporting the best prompt is *p*-hacking.**

claude-sonnet-5 via Claude Code 2.1.278 · 2026-09-20 · prompt 846fe144

Source: This course's own recorded run: 389 headless calls on the frozen statement corpus, 2026-09-20.

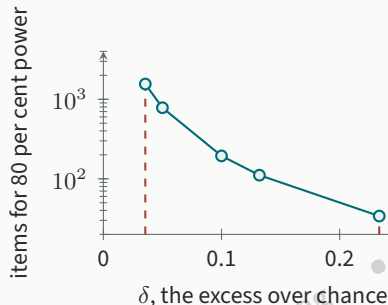
Primacy: The Number, and the Direction



- ▶ On a task whose answer should be fifty-fifty, the first option is chosen 63.21 per cent of the time, and saying the order is random does not remove it.
- ▶ Pooling over the full factorial of 32 architectures returns 50.01 per cent: randomisation, not exhortation.
- ▶ **The direction surprises people:** *symbol* labels give 73.35 per cent against 53.55 for letters, and reviews often print it reversed.

Source: Brucks & Toubia, *PLOS ONE* 20(4):e0319159 (2025), full-factorial design.

The Power Calculation That Settles the Robustness Check



- ▶ Let δ be how far the first-option rate sits above chance. Two-sided at 5 per cent, 80 per cent power:
- ▶ Letter labels, $\delta = 0.0355$: 1,555 items. Symbol labels, $\delta = 0.2335$: 34. A 46-fold difference from formatting alone.
- ▶ On 50 items the power against the smaller effect is 0.10.
- ▶ **So “no order effect on 50 items” is evidence of nothing.**

Source: Miao & Feng, *AI for Economists*, ch. 14, design calculation calibrated on Brucks & Toubia (2025).

Three Design Principles

one
boundary
examples

the dividing
line, not
the centre

two
reason,
then answer

a trace you
can audit

three
call out, do
not recall

external facts
beat stale
memory

- ▶ **Put the boundary in the examples:** demonstrate the hard cases near the dividing line, not the clearest positives.
- ▶ **Reason first, answer second:** a sentence of justification before the label lowers boundary errors and leaves a trace.
- ▶ **Call out rather than recall:** retrieval replaces the stale memory that fabricates.
- ▶ **The tool environment is part of $\mathcal{P}$, so disclose it with the prompt.**

Source: Brown et al. (2020); Wei et al. (2022); Kojima et al. (2022); Yao et al. (2023).

Temperature, Drift, and Four Things to Report

- ▶ **Temperature has two opposite uses.** For annotation, zero: one input, one measure. For simulation, zero is harmful — every simulated subject answers identically.
- ▶ For simulation, treat sampling as Monte Carlo randomisation: draw R times and report the distribution. The book marks this as its own recommendation, not a settled finding.
- ▶ **Version drift:** a named service changes behaviour between releases and you are not told, so $f_{v_1}(\mathbf{x}) \neq f_{v_2}(\mathbf{x})$.
- ▶ **Four minimums:** three wordings on boundary cases; order randomised between items; the version frozen and dated; one independent open-weight replication.
- ▶ **Say why you chose what you chose — one prompt at two temperatures is two instruments.**

Source: Chen, Zaharia & Zou, *Harvard Data Science Review* (2024); Miao & Feng, ch. 14, “最低标准”.

Session 5.5

Verification, Reproduction, and the Record

What the screening can and cannot see, and what has to be written down



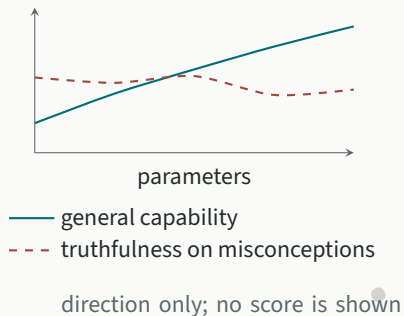
Three Hallucinations, Three Error Structures

type	what it looks like	error structure	what checks it
factual	the assertion is wrong, the language is not	broad signal noise, varying with the domain	sample, and trace the high-stakes ones
citation	author, year, journal and title all conventional	variable substitution: you cite the model's prior	trace each one to the original
numerical	the magnitude is right, the value is wrong	non-classical: correlated with the truth	a range check passes, so go to the raw data

- ▶ **A fabricated citation is not noise, and a range check passes on a fabricated number because the model knows the order of magnitude.**

Source: Miao & Feng, *AI for Economists*, ch. 14, table of the three hallucination types.

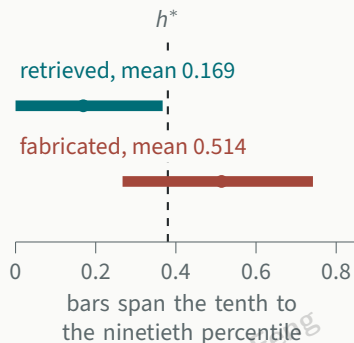
Larger Is Not More Truthful — and What That Benchmark Does Not Say



- ▶ **Inverse scaling:** as parameters grew, general capability rose while truthfulness on common misconceptions did not. Some larger models did worse.
- ▶ **What it does not say:** the benchmark is built on popular misconceptions and does not study citation fabrication. “Bigger models invent more citations” is not that finding.
- ▶ **So never read fluency as reliability: a precise detail stated with certainty raises the checking priority.**

Source: Lin, Hilton & Evans, *ACL* (2022); Miao & Feng, ch. 14, the footnote on inverse scaling.

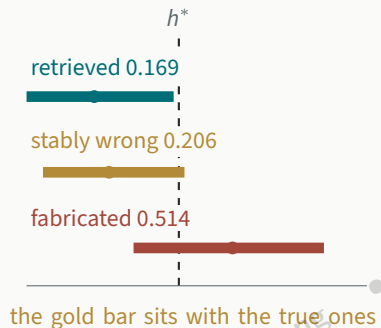
Self-Consistency Screening



- ▶ Sample one prompt R times at non-zero temperature and score the disagreement, $\hat{h} = 1 - \frac{1}{R-1} \sum_{j \geq 2} \text{sim}(s_1, s_j)$.
- ▶ Fabricated statements average 0.514, retrieved ones 0.169. Not a clean split: 0.267 sits below 0.367.
- ▶ At $h^* = 0.38$, precision 0.745 and recall 0.755: 24.5 per cent of fabrications get through.
- ▶ **A larger R steadies the score rather than separating the two, and $R = 8$ is where it stops paying.**

Source: Manakul, Liusie & Gales, *EMNLP* (2023); the book's own simulation on 600 claims, ch. 14.

What the Screening Structurally Cannot See



- ▶ A stably held wrong fact makes every sample agree, so $\hat{h}$ is as low as for a true one: 88.8 per cent of them go unflagged.
- ▶ **It measures the variance of fabrication, not truth.** It orders human checking; it is not a test.
- ▶ This course's run says it from the other side: the byte-identical prompt agreed with itself 81 per cent of the time, so the variance here is sampling noise.
- ▶ **Only a source catches a stable error.**

claude-sonnet-5 via Claude Code 2.1.278 · 2026-09-20 · prompt 846fe144

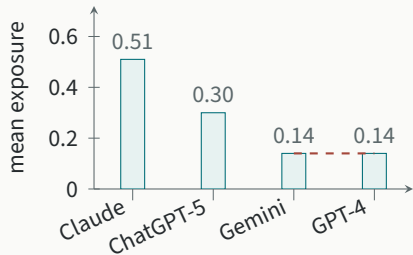
Source: Miao & Feng, *AI for Economists*, ch. 14, “自洽性筛查”; Lewis et al. (2020) for the retrieval route.

Drift, Hardware, and When Averaging Runs Stops Working

- ▶ Two versions of one service are two instruments, $f_{v_1}(\mathbf{x}) \neq f_{v_2}(\mathbf{x})$, under a single name. Record the vendor, the exact version, the system prompt, the sampling parameters and the dates.
- ▶ Even with the version fixed and the temperature at zero, bit-exact reproduction is not promised: a parallel reduction reorders the floating-point sums, and addition is not associative.
- ▶ A last-digit difference can reorder the softmax and change which token wins, so set the standard where it belongs: the *conclusion* reproduces, not the text.
- ▶ Averaging R calls divides the variance by about R — *if* the errors are independent.
- ▶ **If the error comes from $\mathcal{P}$, every call makes it, and a larger R computes a biased estimate more precisely.**

Source: Miao & Feng, *AI for Economists*, ch. 14, “结果复现规范” and the box on hardware non-determinism.

Three Models, One Score Sheet



the 2026 model matches the older one; against itself a model clears 90 per cent

- ▶ Same rubric, same 18,797 tasks, temperature zero; only the rater changes: exposure runs 0.14 to 0.51, a 3.6-fold range, and κ between models 0.36–0.56.
- ▶ **The sharp point:** one 2026 model reproduces the original GPT-4 mean almost exactly; the provider's own successor more than doubles it.
- ▶ **Downstream, coefficients vary 2.4-fold, and the one significant county estimate is a single 2026 annotator's.**

Source: Yin, Vu & Persico, NBER Working Paper 35110 (April 2026), Tables 1 and 2.

Three Generations of One Instrument

generation	the instrument	re-run next year	what you can audit
ability mapping	a hand-curated map from benchmarks to task abilities	identical	the raters' notes
patent overlap	term-frequency overlap of patents with task text	identical	the patent and the task
model as rater	one model call per task, on the published rubric	a different number	the stated reason

- ▶ The first two treat the instrument as a *document*, and documents do not drift.
- ▶ **The third treats it as a model call, and a model call has a version.**

Source: Felten, Raj & Seamans (2018, 2021); Webb (2020); Eloundou, Manning, Mishkin & Rock (2023).

When Not to Reach for a Model as Rater

hand-code a gold set,
then fit a plain baseline

clears $\kappa \geq 0.75$ and
beats the baseline: use it

no clear margin:
ship the baseline

in between: ensemble,
and adjudicate by hand

a baseline has no version to drift

- ▶ Hand-code a gold standard, then fit a plain baseline — term frequencies and a logistic regression — and report its κ against that gold.
- ▶ Run the model in parallel. If it does not beat the baseline by a real margin, the baseline wins.
- ▶ **Name the scheme whenever a κ gets an adjective:** the $\kappa \geq 0.75$ threshold is Fleiss's, Landis and Koch have no such cut, and a third scheme is in circulation.

Source: Fleiss (1981); Landis & Koch, *Biometrics* 33(1):159–174 (1977); McHugh, *Biochemia Medica* (2012).

What the Policy Requires, What the Book Recommends

journal policy

four obligations, plus
one expectation about
the reviewer's reading

say in the paper which is which

the book's checklist

nine further items, and the
book calls them its own

- ▶ The policy's artificial-intelligence section runs to five sentences on a page carrying no date: no AI author; sole accountability with fact-checking; a brief description at submission; disclosure in review, and only tools that will not train on the manuscript.
- ▶ It is not a ban on reviewers; the book counts four clauses, leaving out the sentence on the reviewer's own reading.
- ▶ **Session 6 carries this into multi-step chains, where the per-step log becomes the record.**

Source: American Economic Association, *AER* editorial policy, "Use of Artificial Intelligence", read 2026-09-20; Miao & Feng, ch. 14.

Session 5.6

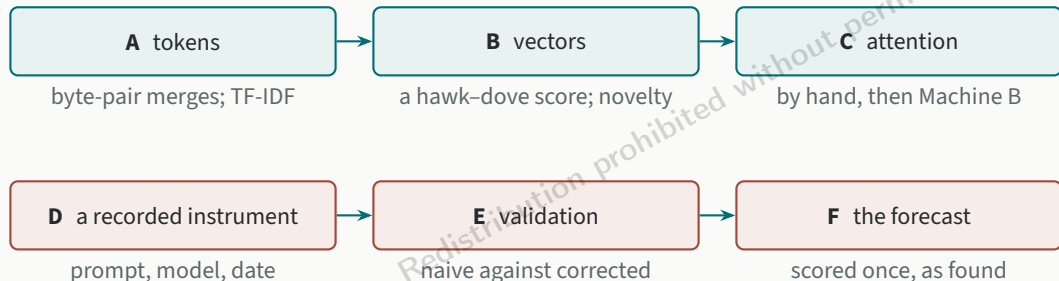
Text Signals and the Bond-Yield Forecast

Four signals, one frozen baseline, one scoring of the test window, and the result as it came out



Zhigang Feng

Six Blocks, and What Each One Has to Clear



- ▶ Blocks A to D build four signals out of one corpus; E asks what a validation set buys; F scores Session 2's forecast once.
- ▶ **Block F's design was hashed before its code existed. No gate depends on text helping.**

Source: The lab, Session 5, blocks A to F; the pre-registered protocol for block F, hashed 2026-09-20.

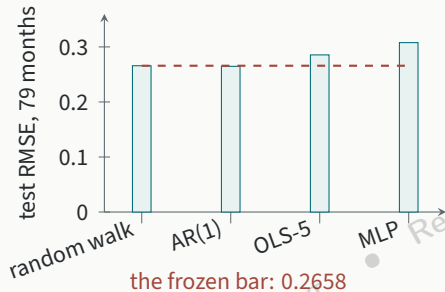
The Rules of the Lab

Block	Runs live	Recorded, and quoted on these slides
A	200 merges; TF-IDF on 149 statements	the same, and the segmentation table
B	co-occurrence, PPMI, 100 dimensions	the same, and the full validation table
C	attention by hand; 60 training steps	Machine B: maps, curves, three temperatures
D	reads a recorded score file	245 statements labelled, 48 four ways
E	the chapter's own scripts	the same, and a sweep over validation sizes
F	the validation window	the test window, scored once against the file

- ▶ Offline, on a CPU, from a clean kernel, fixed seeds. Imports: numpy, pandas, torch.
- ▶ **Every number on these slides is in a recorded file the notebook prints beside its own.**

Source: The lab, Session 5; recorded run of 2026-09-20.

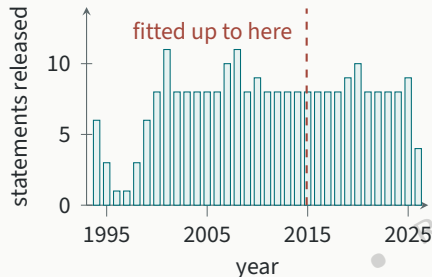
Session 2's Frozen Baseline, Restated in Full



- ▶ One month ahead, the ten-year yield, 79 months. Session 2's three reproducible rows must return to four decimals, or the run stops.
- ▶ The MLP row is two objects: the published 0.3078 averages ten per-seed errors; the mean forecast scores 0.2956.
- ▶ **The bar is the random walk, 0.2658 — AR(1) all but ties it — and it is a very low-variance forecast.**

Source: Session 2's frozen baseline file, scored over 79 months; Deck 2.4.

The Corpus, and the Window Everything Is Fitted On



- ▶ 245 statements, 1994-02-04 to 2026-06-17; 149 fall in the training window, which ends 2014-12-31.
- ▶ Every fitted object is fitted there; 2015–2019 chooses penalties; the test window enters no choice.
- ▶ **A month's signal is the last kept statement dated on or before that month's last observation, and no later document reaches the row.**

Source: The lab's frozen corpus manifest; the pre-registered protocol, information rule.

Five Releases That Are Not Policy Decisions

Date	What it is	Why it is not a policy decision
2007-08-10	a market-functioning notice	the Desk supplies reserves; the target is unchanged
2008-03-11	a facility announcement	a lending facility and swap lines, with four other banks
2010-05-09	a swap-line renewal	a balance-sheet action, no rate decision, no stance
2019-10-11	a reserve-management notice	bill purchases and repos, called purely technical
2025-08-22	a framework review	revisions to the longer-run goals; no view on the stance

- ▶ 7 months carry two or three releases; the last kept one wins, and 240 remain.
- ▶ **The list was read, not sampled, and it sits inside the hashed protocol.**

Source: The pre-registered protocol, exclusion set; the Committee's own release texts.

Session 5.6

Four Signals Out of One Corpus

Blocks A to D: tokens, vectors, attention, and an instrument that reads



Block A: Byte-Pair Merges, Learned by Hand

the first merges, in order

e+_ i+n t+h th+_e_

a word the window uses often

raising
ra | is | ing_

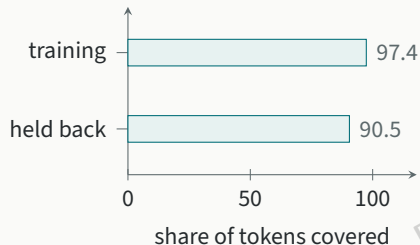
a word it never uses

reflationary
re | f | l | a | tion | ar | y_

- ▶ 200 merges on the 149 training statements: count adjacent pairs, merge the commonest, repeat.
- ▶ The merges follow this corpus's frequencies, not English's.
- ▶ “Raising” takes 3 pieces; “reflationary” is not in the 1,628-word vocabulary and takes 7.
- ▶ **Change the window and the same word splits differently.**

Source: Sennrich, Haddow & Birch (2016); Miao & Feng, *AI for Economists*, ch. 13; the lab, block A.

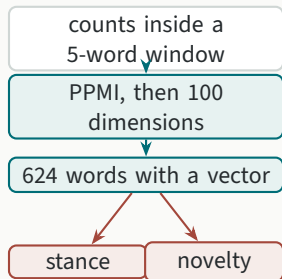
Block A: TF-IDF, Fitted Where It Is Allowed To Be



- ▶ A term is kept if it appears in at least 2 training statements: 1,163 columns, and 85.5 per cent of the 245-row matrix is zero.
- ▶ Coverage falls from 97.4 per cent of training tokens to 90.5 per cent of held-back tokens: the vocabulary ages.
- ▶ **Fitting on all 245 statements would raise the second bar and destroy the experiment.**

Source: The lab, block A, TF-IDF fitted on the 149 statements of the training window.

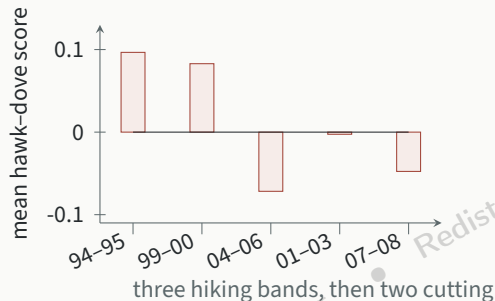
Block B: A Hawk–Dove Axis and a Novelty Index



- ▶ Count which words occur near which, weight by pointwise mutual information, truncate to 100 dimensions.
- ▶ **Stance** is closeness to 5 hawkish seeds minus closeness to 4 dovish ones. **Novelty** is one minus the cosine between consecutive statements, and is zero in a month with no meeting.
- ▶ **Both are fitted inside the training window — the property the next page puts on trial.**

Source: The lab, block B; Miao & Feng, *AI for Economists*, ch. 13, on distributional representations.

Block B: The Embedding Score Fails Its Validation



- ▶ The outside standard is what the Committee did next: hawkish scores on bands of increases, dovish on cuts.
- ▶ The 2004–06 hiking cycle, 17 statements, scores -0.072 — the most dovish stretch in the sample — while two other hiking bands score 0.097 and 0.083.
- ▶ **Rank correlation with the next 3 meetings' path is $+0.07$. That is the result, not a bug to be repaired.**

Source: The lab, block B, validated on the training window against the subsequent policy path.

Block B: The Rule That Passes, and What It Actually Is

signal	hiking bands	cutting bands	rank corr.	verdict
embedding hawk–dove	0.036	−0.025	+0.07	fails
decision sentence	0.810	−0.568	+0.68	passes
novelty index	0.189	0.180	−0.06	fails

- ▶ The rule that passes is two regular expressions — 7 tightening verbs, 9 easing verbs — read on the sentence that says what the Committee decided.
- ▶ **That sentence restates the funds-rate change the panel already carries. A signal can pass validation by re-deriving a number you have.**

Source: The lab, block B, three signals scored on 149 statements of the training window.

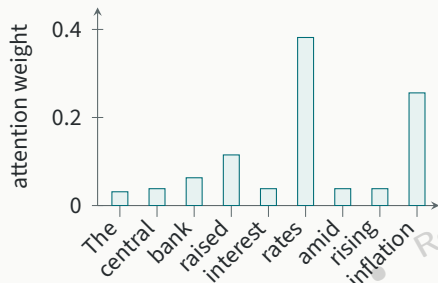
Session 5.6

Inside the Model, and Beside It

Block C: attention by hand, then Machine B. Block D: a score whose information set you cannot date



Block C: Attention by Hand, Against the Book



- ▶ The chapter stipulates 9 scores for one sentence, with “interest” as the query, and prints the weights.
- ▶ The lab recomputes the softmax: the exponentials sum to 26.13, “rates” takes 0.382 and “inflation” 0.256, and all 8 printed figures agree.
- ▶ **Doing the arithmetic is how you learn the example was stipulated, not produced by a model.**

Source: Miao & Feng, *AI for Economists*, ch. 13, worked attention example; the lab, block C.

Block C: Machine B's Attention, Layer by Layer

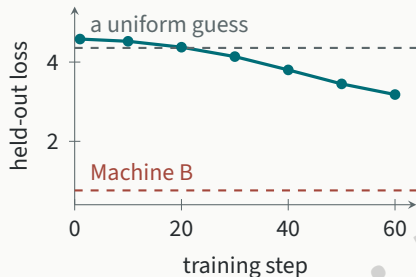
	head 1	head 2	head 3	head 4
layer 1	3.27	3.25	3.30	3.10
layer 2	2.04	1.68	1.00	1.74
layer 3	1.49	1.57	1.11	1.52

mean row entropy on one sentence; 4.32 is the flattest possible

- ▶ **Machine B** is Session 1's character-level transformer: 631,424 parameters, 3 layers, 4 heads, trained 3,000 steps to a held-out loss of 0.761.
- ▶ Layer 1 stays diffuse in all four heads; by layer 2 one head has collapsed to 1.00.
- ▶ **Nobody asked for a division of labour between heads; training produced it.**

Source: The lab, block C, on Session 1's frozen model and a 75-character sentence.

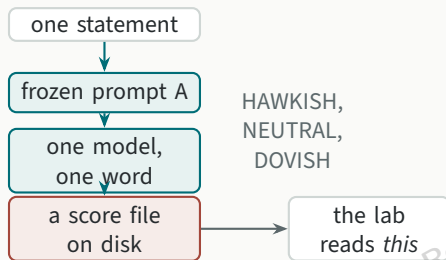
Block C: Sixty Steps, and Three Temperatures



- ▶ Live: 60 steps, about 15 seconds. Loss 4.579 to 3.260, where a uniform guess over 78 characters scores 4.357.
- ▶ Machine B's 3,000 steps take about 18 minutes. Sampled at 0.5, 0.8 and 1.2, its phrasing holds at the lowest and dissolves at the highest.
- ▶ **Fluent phrasing is the cheap part of the objective, not evidence about policy.**

Source: The lab, block C; the live run and Session 1's recorded training curve.

Block D: A Stance Score as a Recorded Instrument

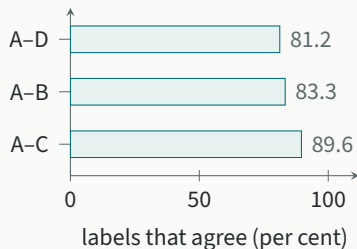


- ▶ 245 statements, one frozen prompt, one word of output mapped to +1, 0, -1: 84 hawkish, 61 neutral, 100 dovish.
- ▶ Every reply is kept with its date, the prompt's hash and the model that answered. The lab never calls a model; it reads a file.
- ▶ **A number a model produced is reproducible only if the run was recorded.**

claude-sonnet-5 through a 2.1.278 harness · 2026-09-20 · prompt A, 846fe144b241d9d1

Source: The lab, block D; the recorded stance run of 2026-09-20, 389 calls in all.

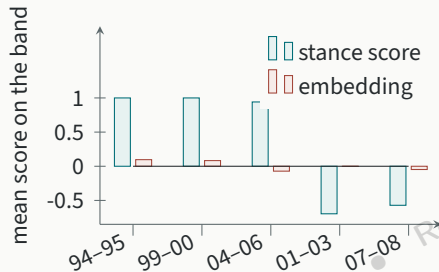
Block D: The Same Prompt, Twice



- ▶ Four variants on the same 48 statements. B rewords the instruction, C reverses the order of the labels, D is A again, character for character.
- ▶ A and D agree on 81.2 per cent: 9 statements change their label when nothing changes. Rewording gives 83.3, reordering 89.6.
- ▶ **Prompt sensitivity cannot be read off a gap smaller than the instrument's own noise.**

Source: The lab, block D, four prompt variants on a 48-statement subsample.

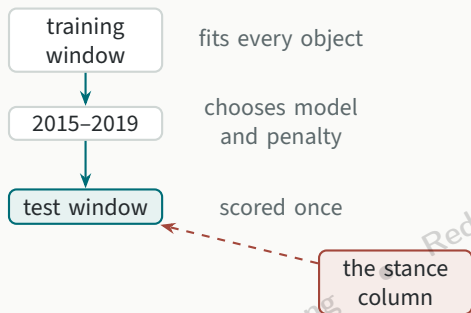
Block D: The Policy Path as an Outside Check



- ▶ Same bands, same training window: hiking 0.980 against cutting -0.634 , and rank correlation $+0.49$ with the subsequent path.
- ▶ On the 2004–06 cycle the embedding read backwards, the stance score reads 0.941 — with no seed list and no fitting.
- ▶ **It is also the one signal whose information set nobody here can date. Hold that until the last pages.**

Source: The lab, blocks B and D, scored on the same 149 statements of the training window.

Block D: Reported Beside the Forecast, Never Inside It



- ▶ Only variant A, run on every statement, may enter a forecast; B, C and D are diagnostics.
- ▶ No model choice and no penalty may depend on the stance column: it is scored beside the structured columns, never pooled into them.
- ▶ **Machine B is excluded by assertion — it was trained on all 245 statements, the test window included.**

Source: The pre-registered protocol, sections on the language-model column and on excluded artefacts.

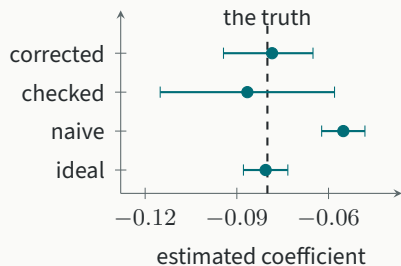
Session 5.6

What Validation Buys, and the Forecast

Blocks E and F: the book's own simulations, then one scoring of the test window



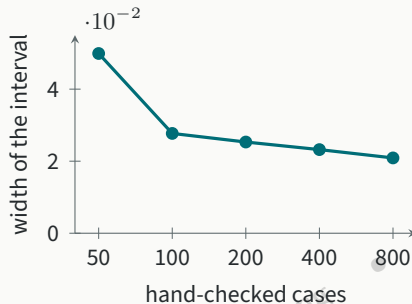
Block E: Naive, Checked-Only, Corrected



- ▶ The chapter's simulation: 3,000 observations, a true coefficient of -0.080 , a label correlated 0.851 with the truth, 200 checked by hand.
- ▶ Treating the label as the truth returns -0.055 , 31.0 per cent of the way to zero, with an interval that excludes it.
- ▶ **Calibrating on the checked cases is unbiased and 48.7 per cent narrower than using them alone.**

Source: Miao & Feng, *AI for Economists*, ch. 14, bias-correction simulation, run unchanged.

Block E: What a Validation Set Buys



- ▶ The interval narrows 49 per cent between fifty hand-checked cases and 200, and a further 17 per cent between 200 and eight hundred.
- ▶ What those cases estimate is the calibration slope: 0.611 at fifty against 0.753 at 200.
- ▶ **Most of what a validation set buys, it buys early. Budget for a few hundred.**

Source: Miao & Feng, *AI for Economists*, ch. 14, bias-correction script; the lab, block E.

Block F: Scored Once, and the Power to Detect a Gain Is Low

250 rows fit

60 choose

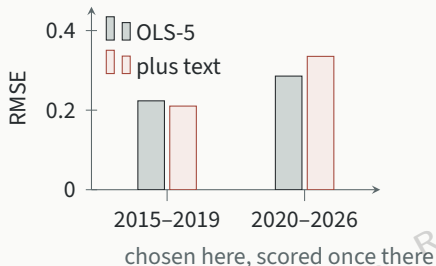
79 rows, scored once

50 of Session 2's training rows pre-date the first statement and are dropped, not filled; the 2025-11-30 gap is inherited and never filled

- ▶ The sample starts at 1994-02-28: no earlier row could carry a text signal. 250 rows fit, 60 choose, 79 are scored once, after every choice was made.
- ▶ All 7 gates are on procedure: hashes, the frozen baselines, the split sizes, the inherited gap, no post-cutoff document in a fitted object.
- ▶ **Say this before the table: with 79 months the power to detect a small gain is low. Not significant is not the same as useless.**

Source: The pre-registered protocol, sample definition and gates; hash 42ae9f04550d6f29, taken 2026-09-20.

Block F: On the Window Where It Was Chosen, Text Helped



- ▶ The model and the ridge penalty were chosen on 2015–2019, and nowhere else. There, text *helps*: 0.2234 against 0.2102, or 5.9 per cent better.
- ▶ On the test window the same pair is 0.2856 against 0.3352: 17.4 per cent worse.
- ▶ **A signal can look good exactly where you chose it. That is what a hashed protocol is for.**

Source: The lab, block F: validation metrics recomputed live, test metrics from the recorded file.

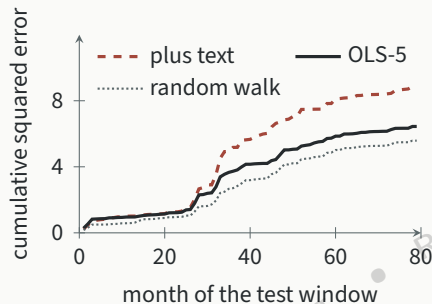
Block F: The Test Window, Scored Once

model	frozen file		re-estimated		plus text	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
random walk	0.2658	0.2097	0.2658	0.2097	–	–
AR(1)	0.2647	0.2089	0.2648	0.2090	–	–
OLS-5	0.2855	0.2255	0.2856	0.2247	0.3352	0.2567
OLS-5 + TF-IDF ridge	0.2855	0.2255	0.2856	0.2247	0.3364	0.2584
text alone	–	–	–	–	0.2685	0.2105
TF-IDF ridge alone	–	–	–	–	0.2675	0.2099
LLM stance column	–	–	–	–	0.2714	0.2146

- ▶ Re-estimating costs 0.2855 to 0.2856; the third column is the text's doing.
- ▶ **Text does not help: added to OLS-5 it is clearly worse, and alone it lands behind the frozen bar, 0.2658.**

Source: The lab, block F, one scoring of 79 test months; Session 2's file in the first pair.

Block F: The Path, and Two Statistics That Agree



- ▶ Diebold–Mariano, Harvey–Leybourne–Newbold corrected: -2.817 and -2.948 ($p = 0.006, 0.004$), both favouring plain OLS-5.
- ▶ Clark–West, built to find such a gain, gives -0.597 and -1.777 and declines to.
- ▶ Text alone against the frozen bar: -1.165 , $p = 0.248$. Indistinguishable.
- ▶ **The text model leads for 6 months of 79 and trails for 73.**

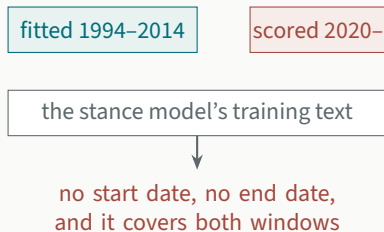
Source: Diebold & Mariano (1995); Harvey, Leybourne & Newbold (1997); Clark & West (2007); the lab, block F.

Where This Can Mislead: Chosen on One Window, Scored on Another

- ▶ The finding is not “text does nothing”. Text looked like it helped on the window where the model was chosen, by 5.9 per cent, and then cost 17.4 per cent on the window scored once.
- ▶ A student who sees only the test column learns that text is useless. One who sees both learns why the protocol file has a hash, and why the hash was taken before the code ran.
- ▶ Four signals, two scopes for the decision rule, several validations and 5 penalties were tried — validation pushed the ridge to 100, shrinking the text almost away, and it still landed behind the bar.
- ▶ Report the arms that failed, or what survives is a selected maximum with fictitious error bars.
- ▶ **The gap between the validation column and the test column is the size of the mistake you would have published.**

Source: The lab, block F, validation and test windows; the pre-registered protocol, gates and reporting rule.

Where This Can Mislead: An Instrument You Cannot Date



- ▶ What the stance model knows about a 2022 statement includes what was written after it, and where the yield went.
- ▶ 55 statements span the 79 test months, 26 of them repeats. The panel is today's vintage; the text is the only real-time part.
- ▶ **The clean signals are those trained inside the training window; the most capable instrument is the one you cannot date.**

Source: The pre-registered protocol, stale-month rule and excluded artefacts; the lab's corpus manifest.

Your Turn, and Where This Corpus Goes Next

- ▶ Pick one. Refit the embedding on all 245 statements and watch the 2004–06 band. Narrow the decision-sentence rule and rerun its validation. Add the stance column to OLS-5 and report both tests.
- ▶ Report the change you made, the validation table before and after, and one sentence on what moved. If a signal passes, say what it correlates with that you already had.
- ▶ What block F answers: on this window, against this bar, text does not improve the forecast, and the two statistics agree. What it does not answer: anything causal, and anything about a window with more meetings in it.
- ▶ **Session 6 takes these same 245 statements as a library to retrieve from, and the question becomes whether it found the right passage and whether you can tell.**

Source: The lab, Session 5, blocks A to F; the corpus carried into Session 6's retrieval lab.